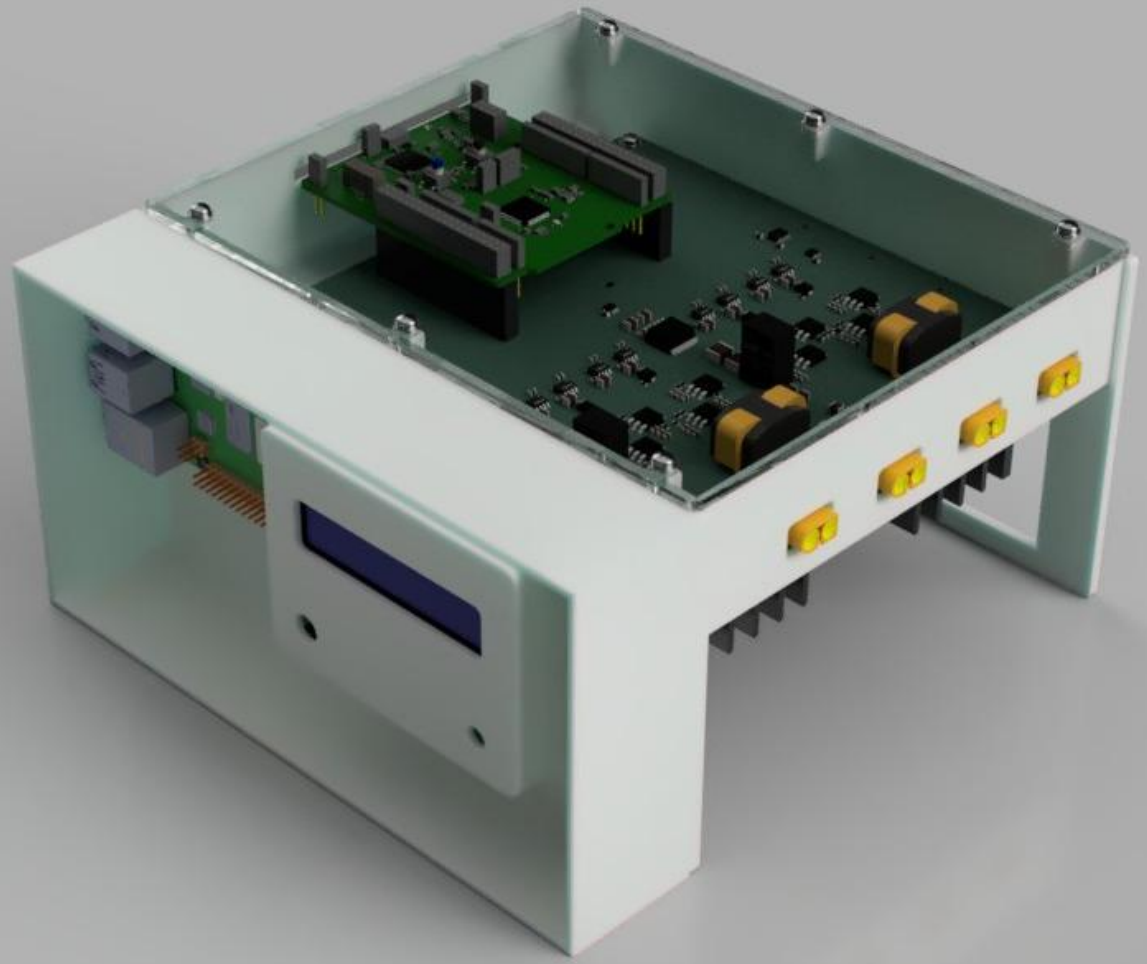
Controlling Differential Power Processors in Solar Powered Applications

By Jared Saunders



Recap – Last Time on Thesis

- Problem
- Research Question
- Existing DPP / MPPT Control Strategies
- Analytical Solution using classical methods
- Partial Simulink Design

* There is way more than that now *

Refined Objectives

- Derive an analytical solution to the dynamics of an interleaved buck-boost converter.
- Design a control system architecture
- Develop a controller board
- Experimental validation on a real photovoltaic system
- Create a self-contained product for controller autonomy

Context- Refined Research Question

Situation

- Solar farms are accounting a large portion of energy production.
- Solar energy is the leading renewable resource.

Key Issues

- Partial shading drops PV performance.
- Untapped energy

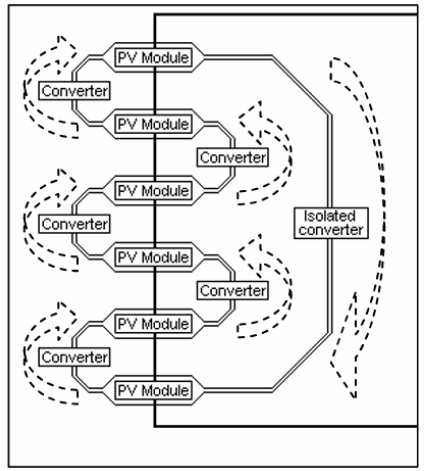
Complication

- Current solutions forego energy

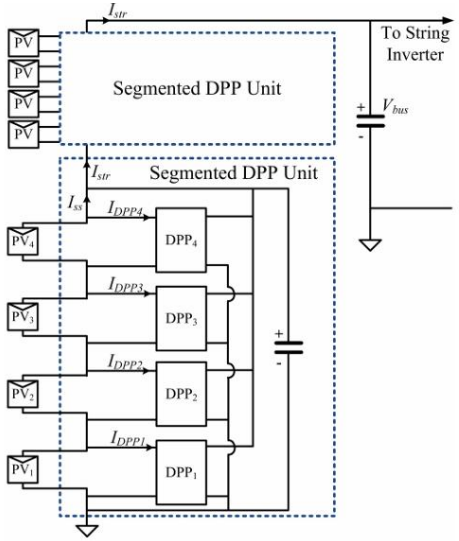
Research Question:

- How to develop a controller that captures the dynamics of multiple interleaved buck-boost converters and commands the currents in a way that reliably extracts the maximum power point despite shading?

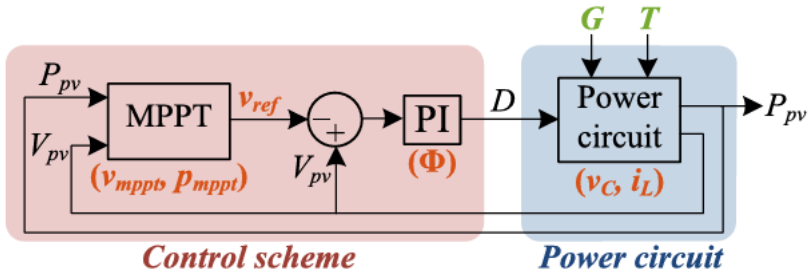
Recap - Existing Partial Shading Mitigation Strategies



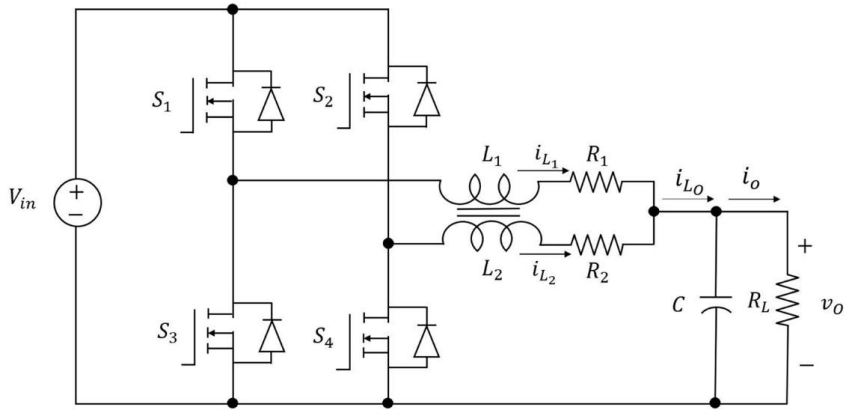
[1] Walker et al.



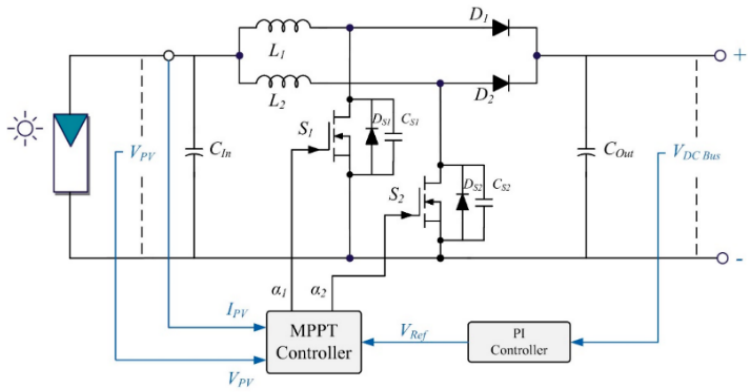
[2] Jeong et al.



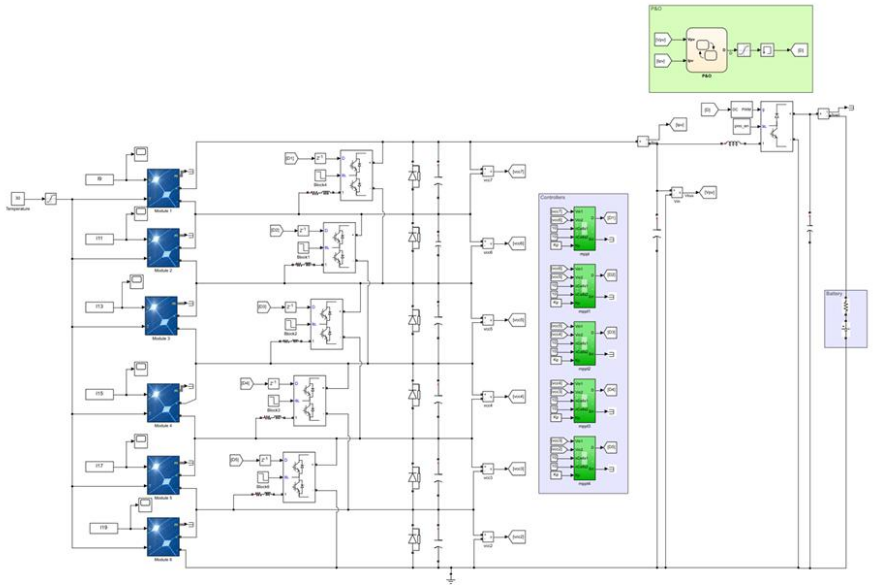
[3] Batzelis et al.



[4] Milasi et al.



[5] Kabalci et al.

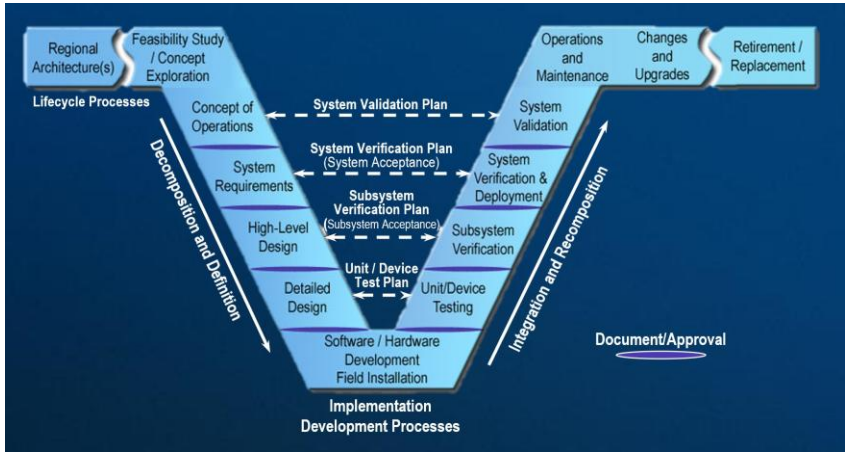


[6] Hui et al.

My Controller Design Pipeline



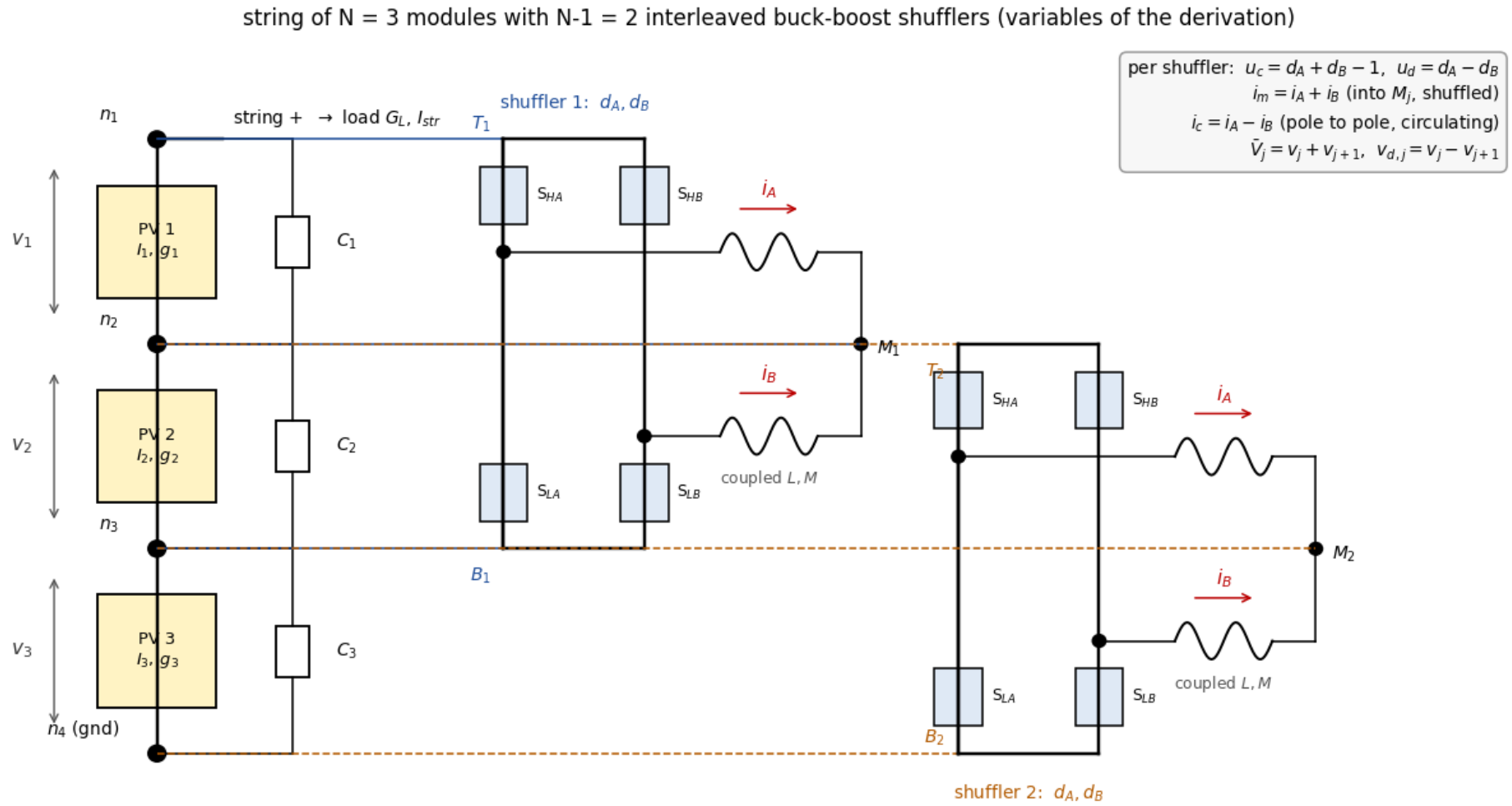
1. Analytical
2. Model-in-the-Loop → Simulink
3. Component Selection (Micro, Sensing, Computing)
4. Software-in-the-Loop → Simulink / QSPICE
5. Theoretical Integration / Gain Refinement → QSPICE
6. Firmware / Software → FreeRTOS / Python
7. Bench Test → PSU input
8. Real Test → PV Panel Input



[1] NUCLEO-G474RE | Product - STMicroelectronics

What's MBSE and what does it mean for designers?

Expanding to N-Modules



System at Equilibrium

Duty Ratio: $\frac{D_j}{1-D_j} = \frac{v_j+1}{v_j}$

Each shuffler carries a weighted sum of all mismatches…

$$T(N)I_m = \delta I; \quad T = tridiag(-0.5,1,-0.5); \quad (\delta I)_j = I_j - I_{j+1}$$

$$T_{jk}^{-1} = \frac{2}{N} min(j,k)\big(N - max(j,k)\big)$$

String Current: $I_{str} = \frac{1}{N}\sum_{k=1}^{N} I_k$

Open Loop Voltage Difference at D = 0.5… $v_{d,j} = -R_m I_{m,j}$

Average N-module Model

SS-Model

$$x = [\phi, i_m, i_c], u = [u_c, u_d], y = [v_d, i_c]$$

Winding Equation 1:

$$(L - M)i_{m,j} = \bar{V}_j u_{c,j} + v_{d,j} - R_m i_{m,j}$$

Winding Equation 2:

$$(L + M)\dot{i_c}, j = \bar{V}_j u_{d,j} - R_c i_{c,j}$$

Nodal Network Capacitance and Voltage

$$C_n\dot{\phi} = -G_n\phi + Si_m + w, \quad v_{d,j} = \phi_j - 2\phi_{j+1} + \phi_{j+2}$$

The **REAL** Analytical Solution

DC Gain Identification Closed Loop:

$$\mathrm{G(N)} = \begin{bmatrix} -(R_m I + Z_e)^{-1} Z_e \, diag(\bar{V}_j) & 0 \\ 0 & \mathrm{diag(\bar{V}_j / R_c)} \end{bmatrix}$$

$$Z_e = -D_v G_n^{-1} S$$

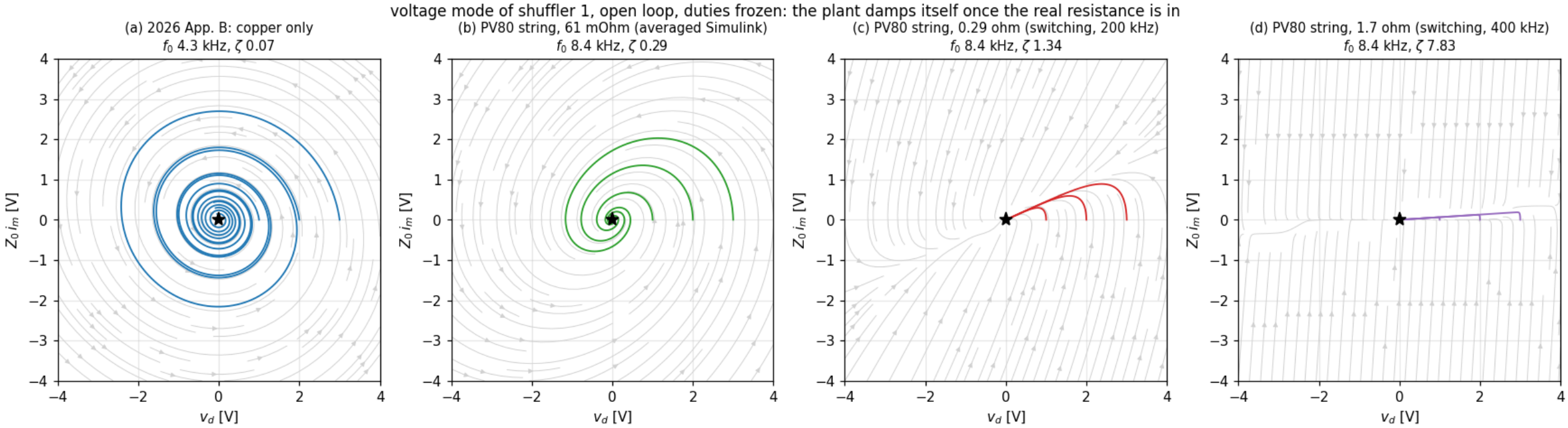
Deployed Controller…

$$M = G^{-1} diag(k_I)$$

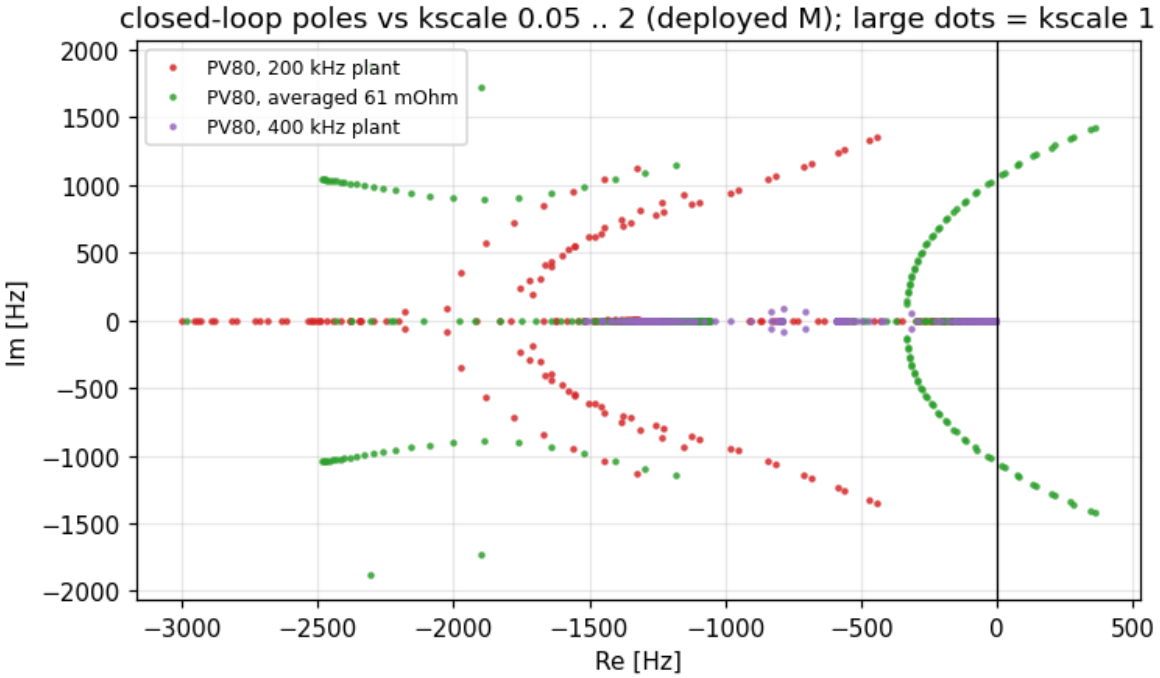
$$u = k\left(M_p e + M\xi\right), \quad \xi i \mathrel{+}= T_s e$$

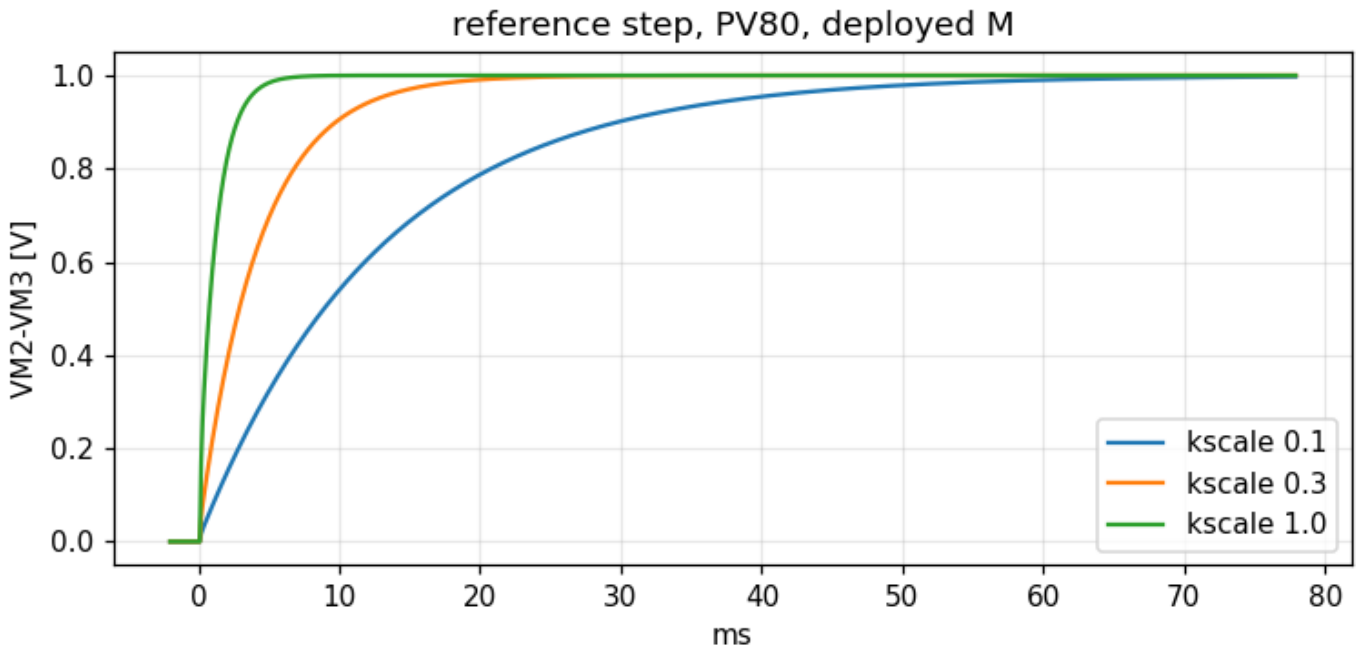
Closed Loop at kscale = 1… $\tau_v = 1.1ms, \quad \tau_i = 0.3ms$

N-Module Phase Portraits

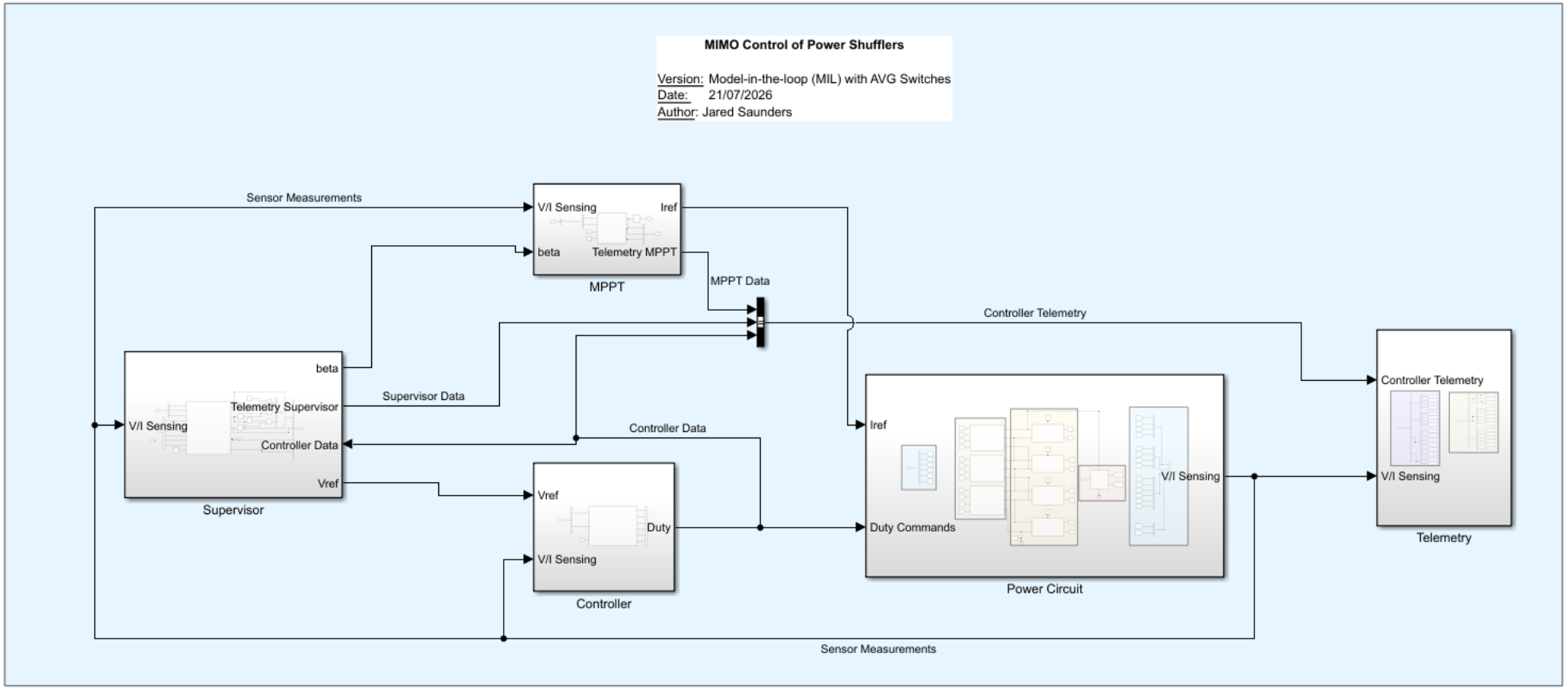


Step Response and Root Locus



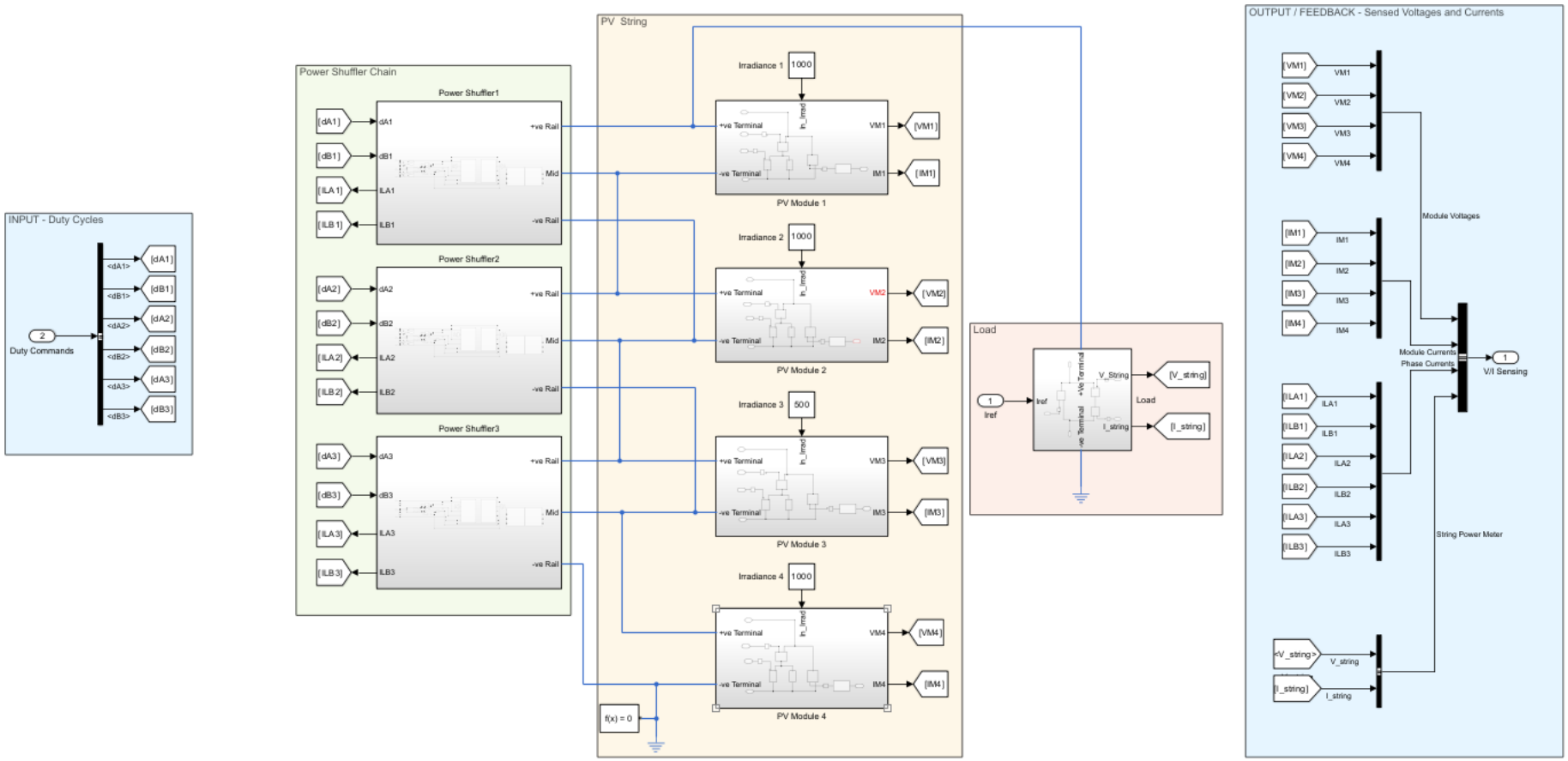


Simulink MIL

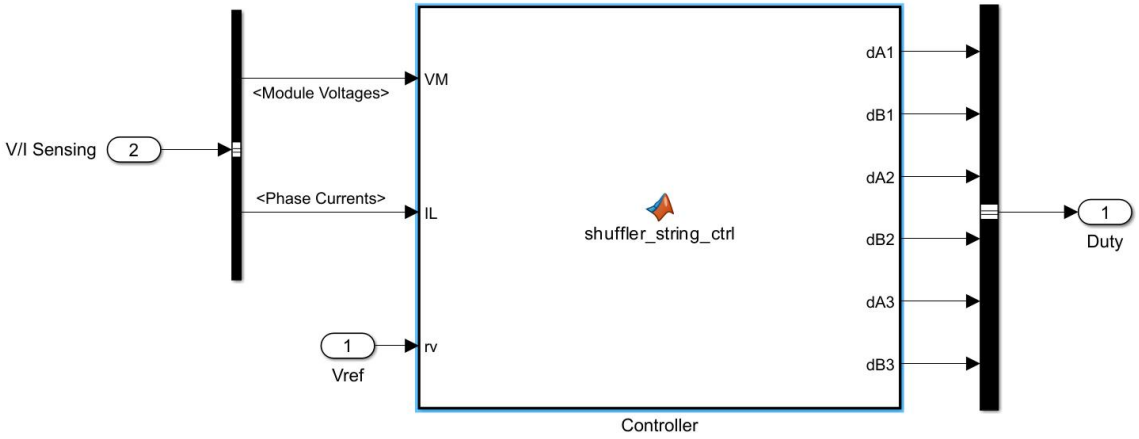


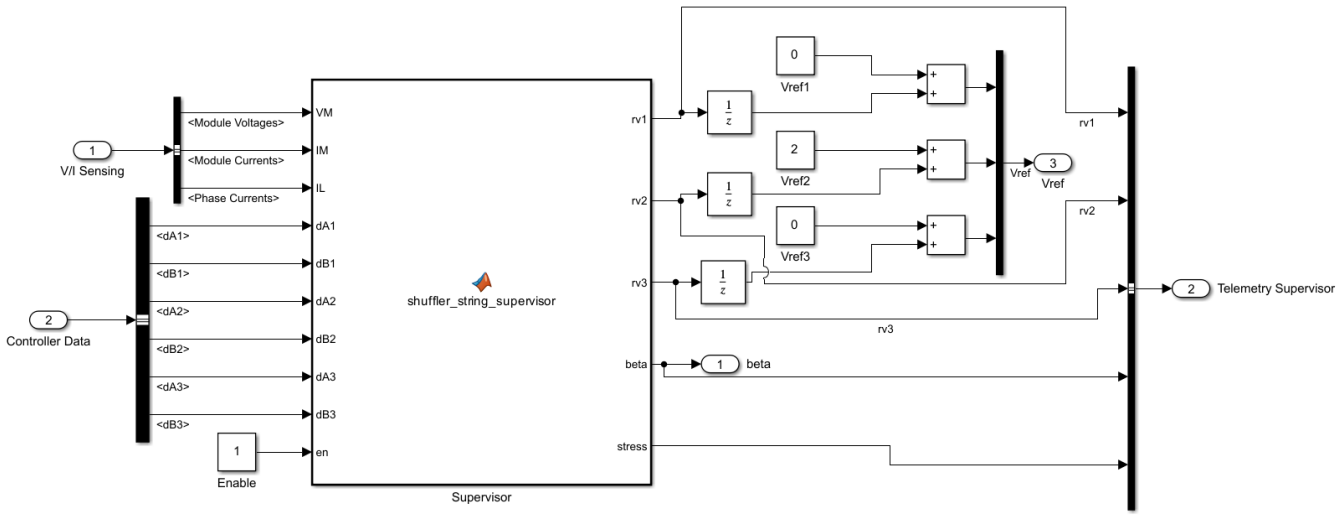
EPISODE II:
ATTACK OF THE SIMULINK

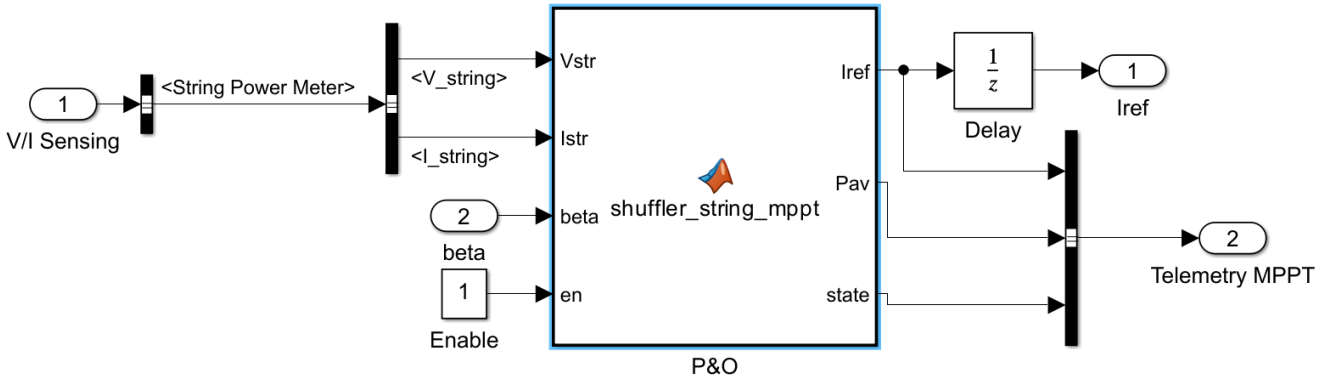
Simulink MIL – Inside the Plant



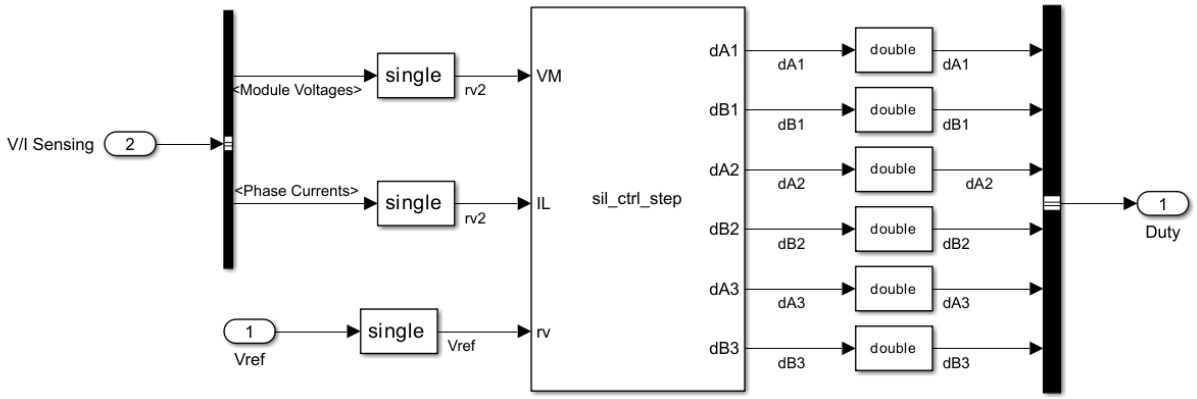
Simulink MIL – The other blocks

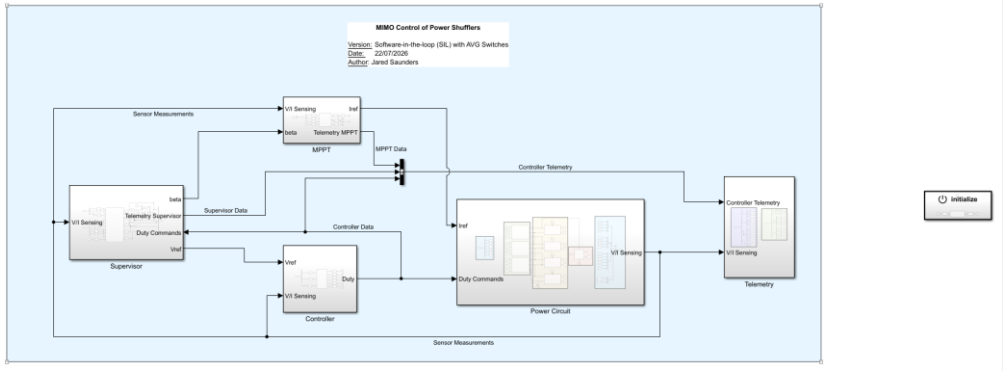


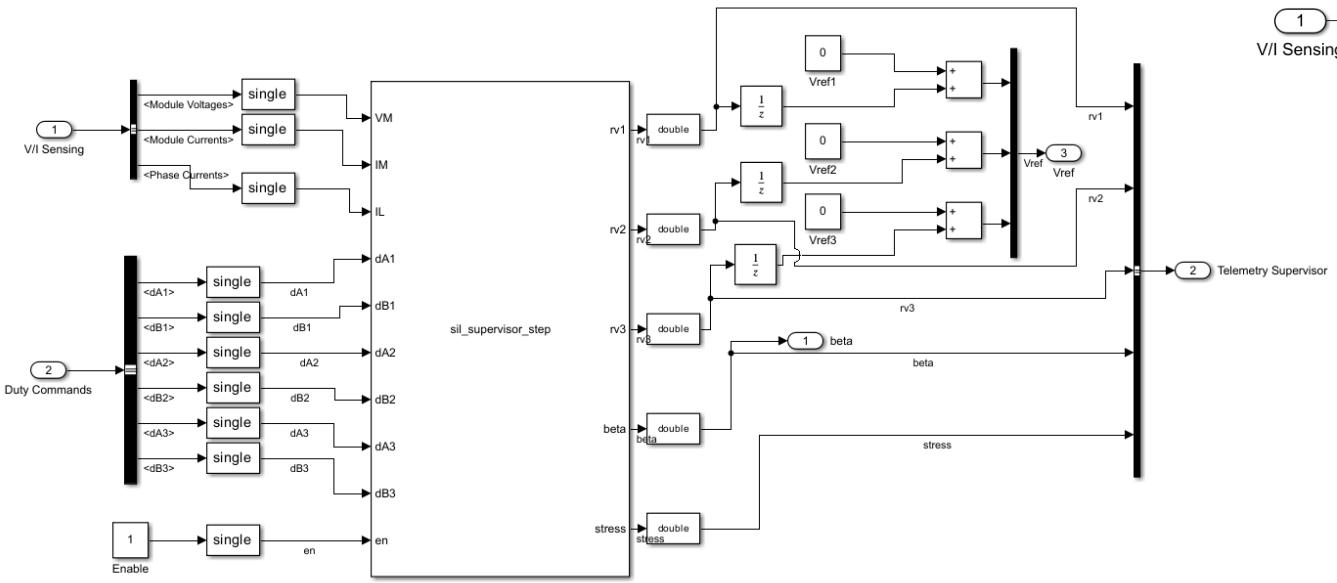


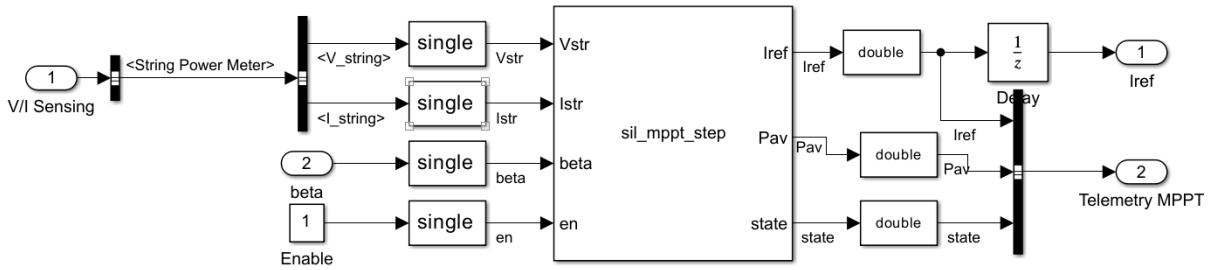


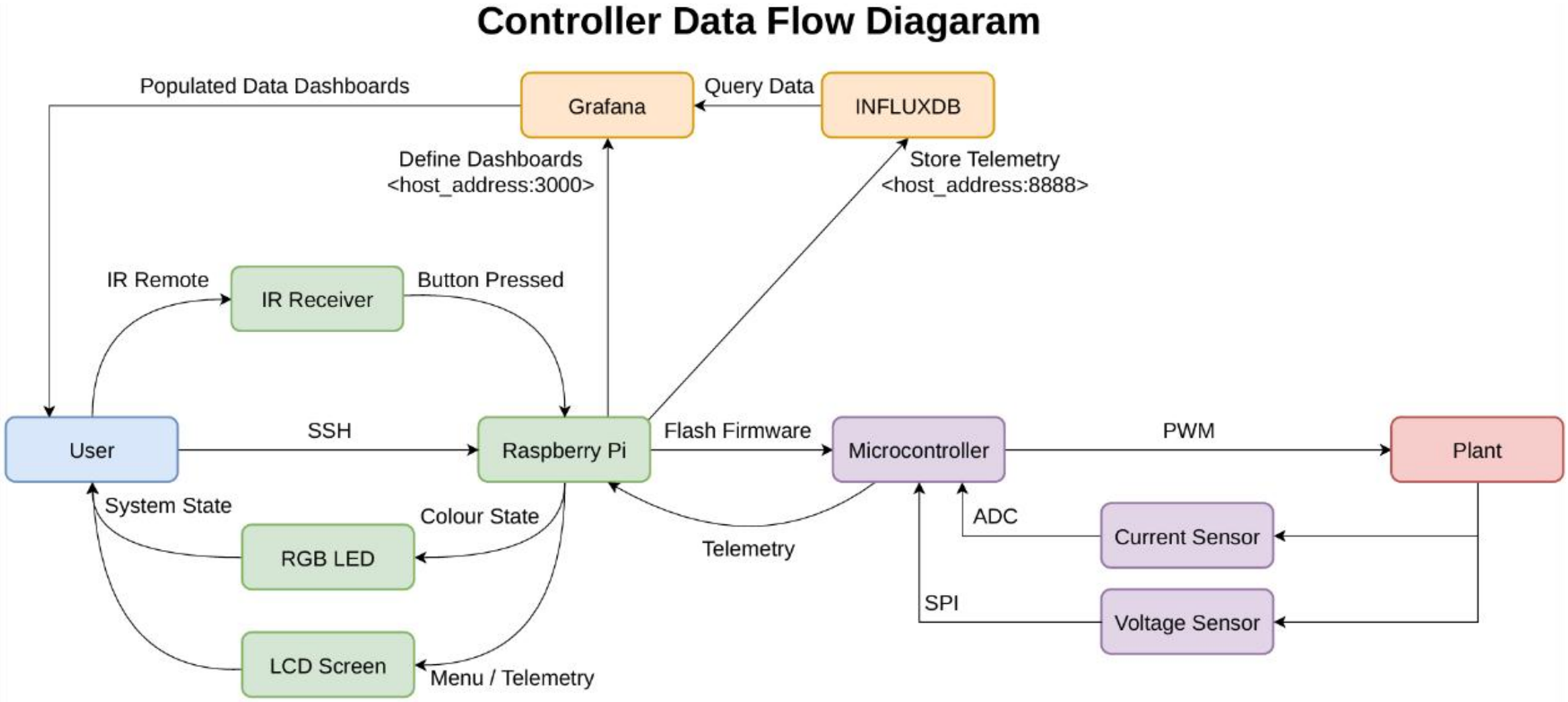
Simulink SIL – MATLAB --> C functions











EPISODE III:
REVENGE OF THE PCB

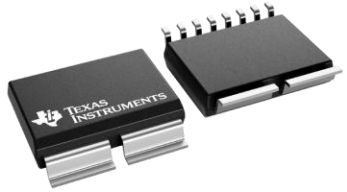
Controller Selection / Routing



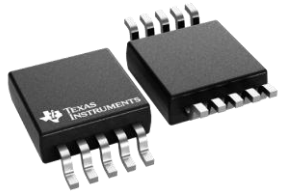
[7]
Microcomputer
(Raspberry Pi 4B)



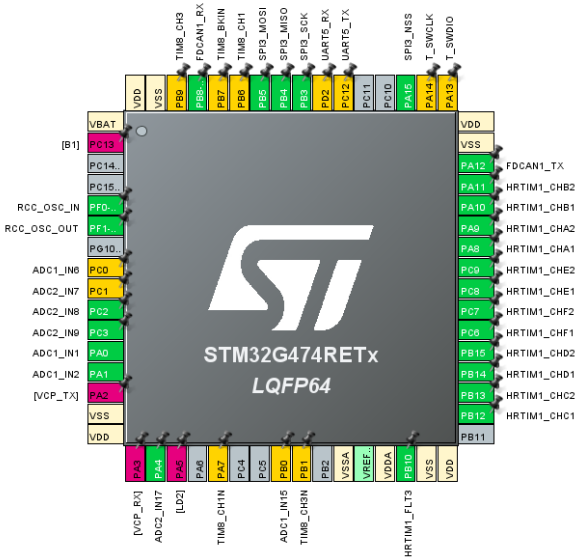
[8]
Microcontroller
(Nucleo G474RE)

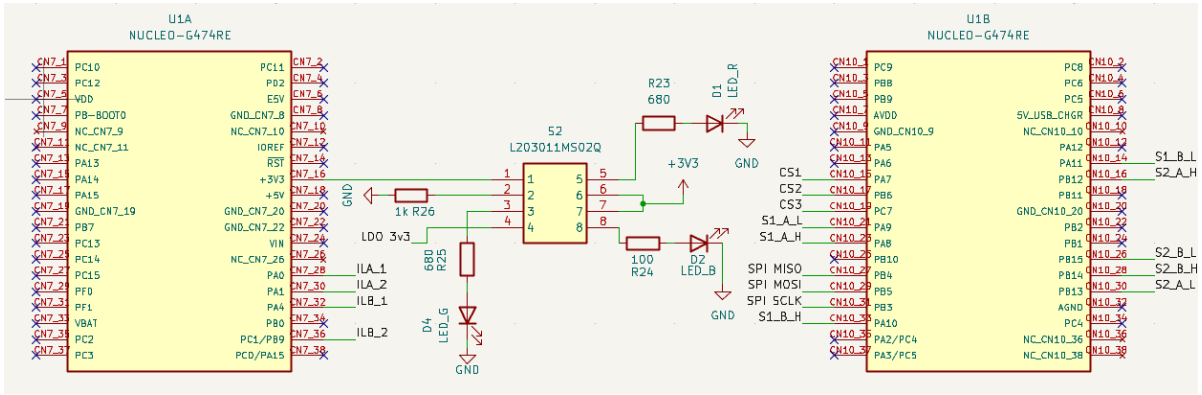


[9]
Hall-Effect Current Sensor
(TMCS1126B4AQDVGRQ1)

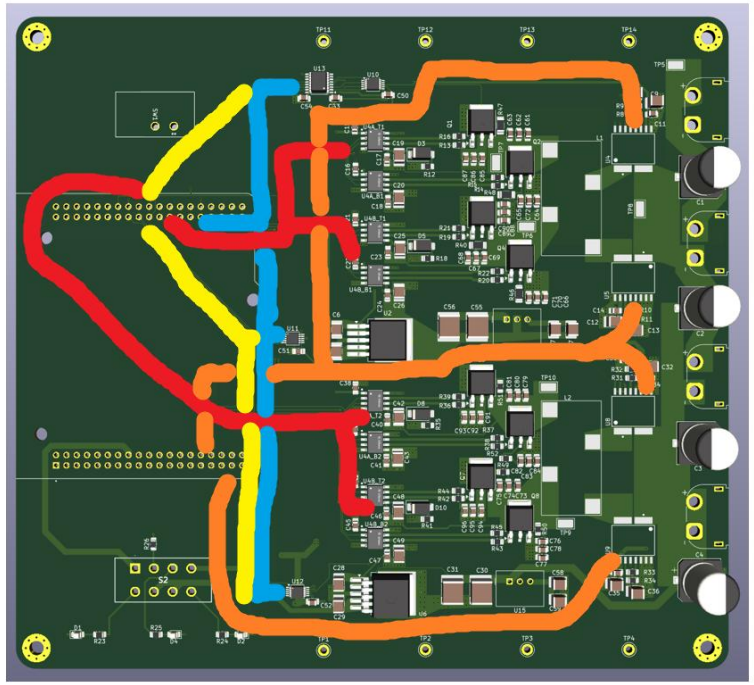


[10]
Power Monitor
(INA239AQDGSRQ1)



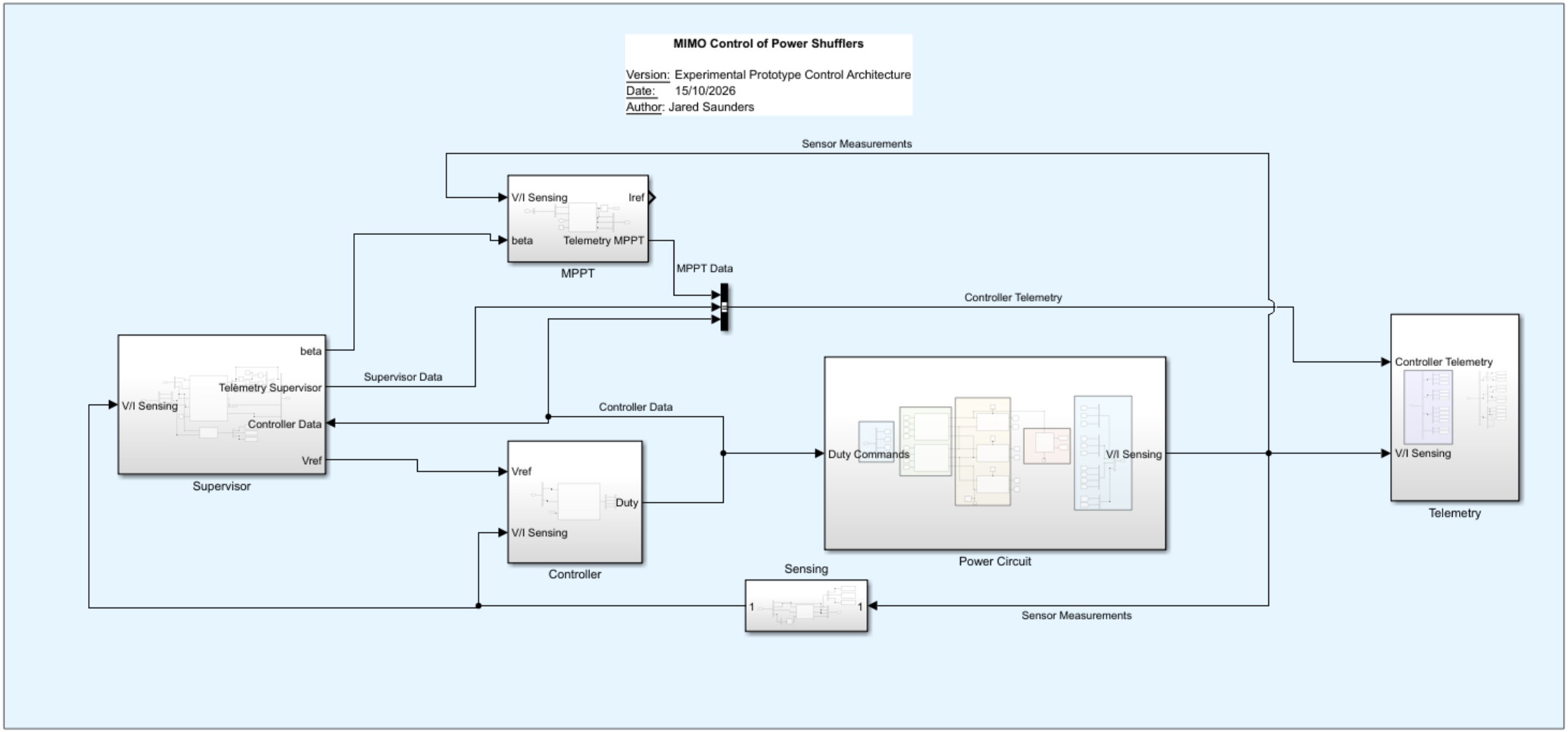


Nucleo Header Pinout



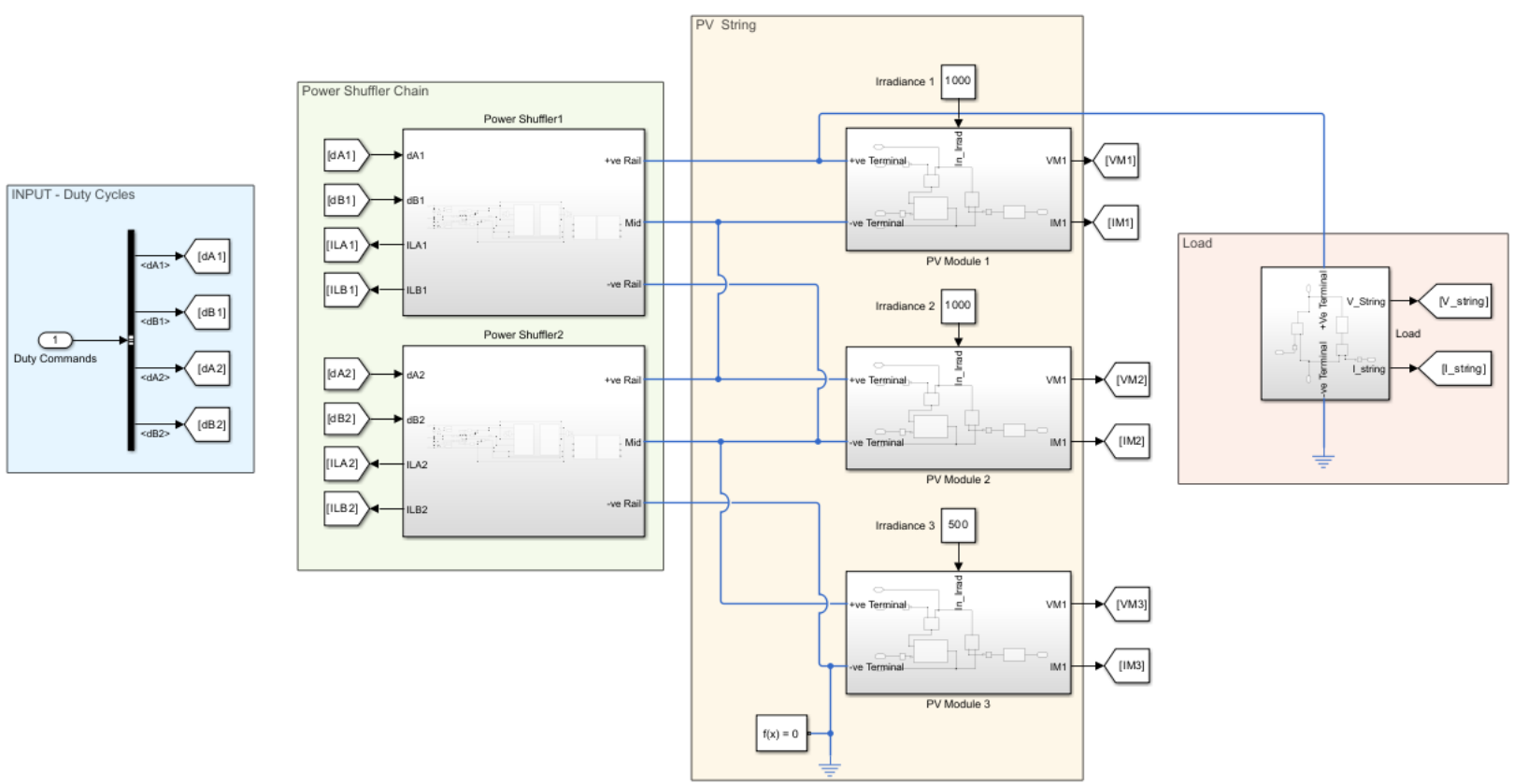
PWM **SPI** **CS** **ADC**

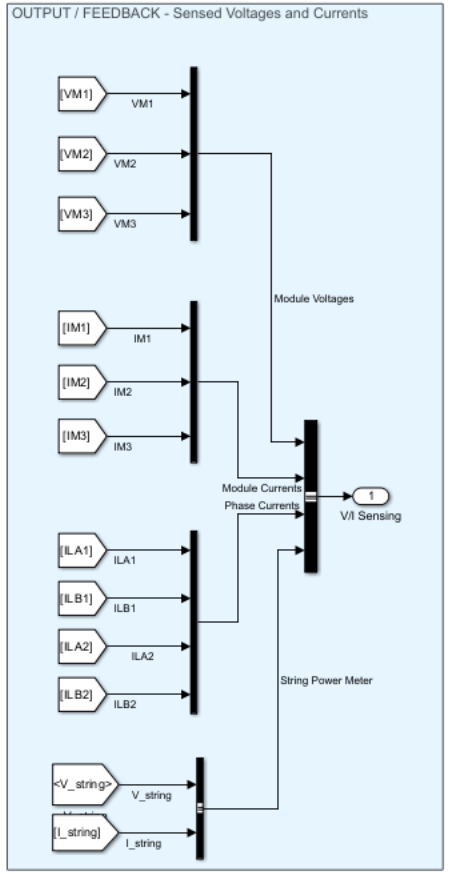
* Routed on Top and Bottom Layers



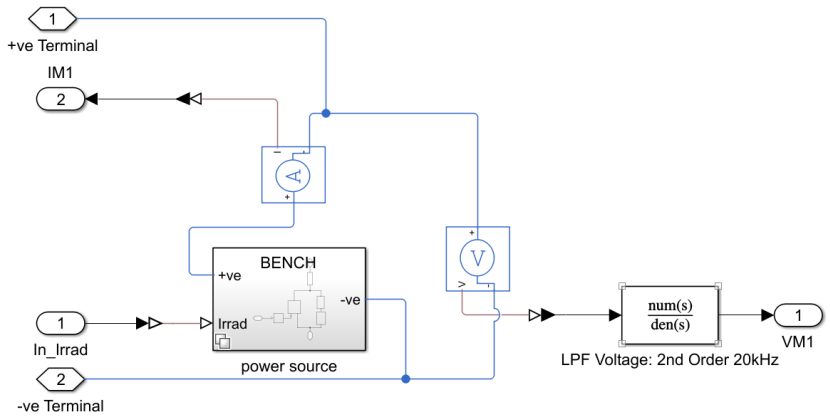
* New sensing block contains a script that inject noise and add delay to the system to represent the current and voltage sensing

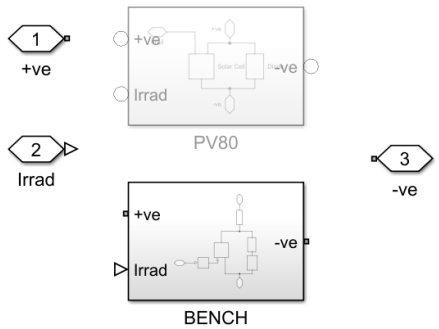
EPISODE IV:
A NEW SIMULINK

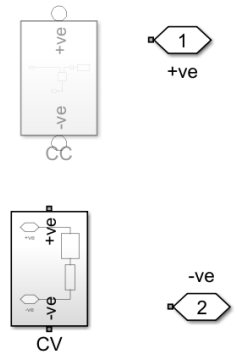


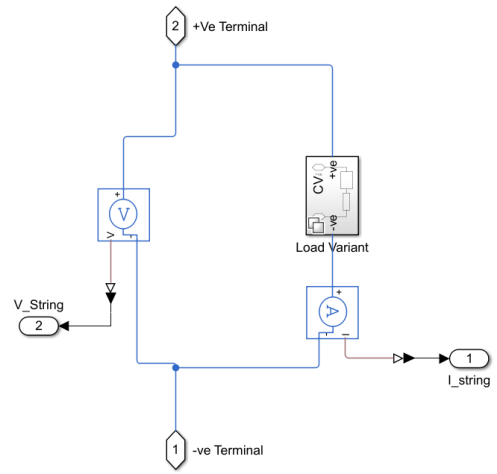


Modified the Plant to match the experiments





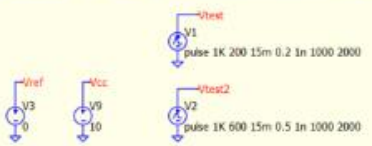




POWER SHUFFLER - the hand-drawn v3 string and converters with the firmware control stack, as hierarchical sheets

Blocks: XS1 / XS2 = Shuffler_1.qsch / Shuffler_2.qsch (the drawn converters with their hall sensors), XC = Controller.qsch (the DLL, its sources,
and two PWM_Modulator.qsch comparator sheets). Right-click a block and open its schematic to look inside; every sheet carries its own notes.
Files that must sit next to this sheet: the four sub-sheets, HalfBridge_Driver.qsch, shuffler_ctrl_dll2.dll, ctrl_M2.txt (MSEL = 2); MyLibrary.lib via the library path.
Nodes: Out = PV1+ at the load, PV2P / PV3P = the INA239 string nodes, T1 / B1 and T2 / M2 / B2 = the converter rails behind the cable and copper inductors L23-L28.
Inside a block, name a current or node with QSPICE's bullet: I(V4•XS1), V(DA1•XC). The PARAMETERS column is the one place to edit.
Generated by Claude's outputs/build_power_shuffler_integrated.py from Buck Boost Full Bridge - v3.qsch (which it never writes); hand edits and the generator do not mix.

SHADE RAMPS V1 / V2 (Vtest, Vtest2: irradiance of module 2 in both strings), Vref and Vcc as drawn

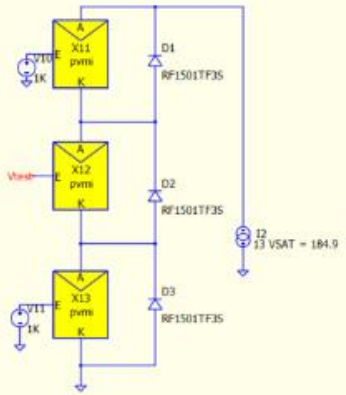


LOAD I1 on Out (13 A ramp, VSAT 137.4 V)

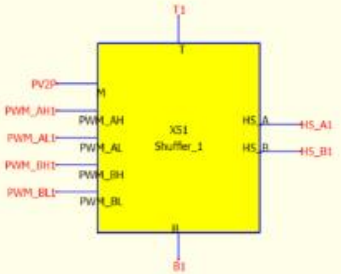


PV STRING - X1-X3 Alko modules, L1-L6 lead inductors, D4-D6 bypass diodes, L23-L28 cable / copper to the converters

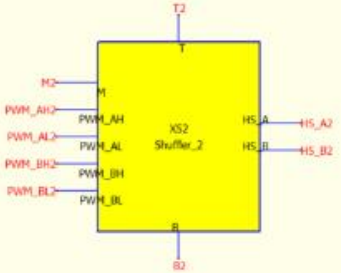
REFERENCE STRING - the same modules and load, never gets a converter (comparison only)



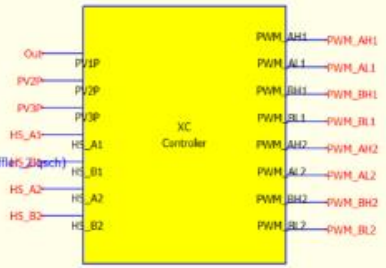
SHUFFLER 1 - converter 1 on PV1+ / PV2+ / PV3+ (Shuffler_1.qsch)



SHUFFLER 2 - converter 2 on PV2+ / PV3+ / GND (Shuffler_2.qsch)

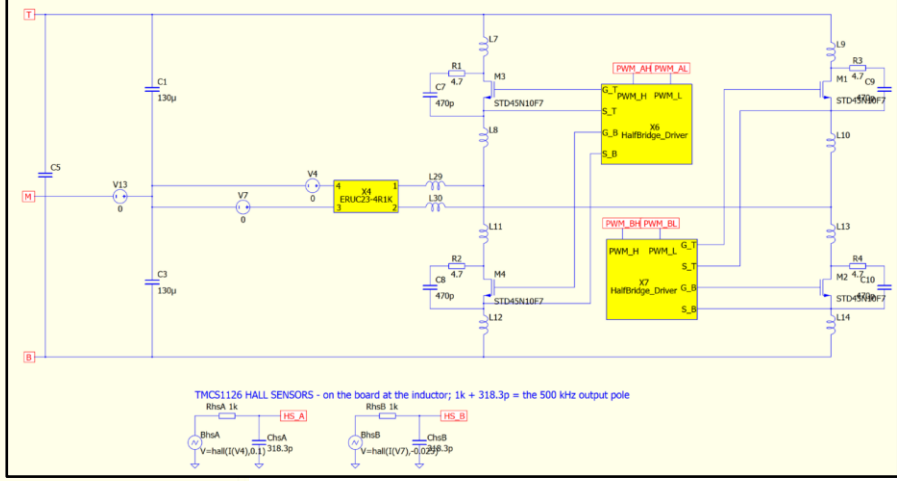


CONTROLLER - firmware control stack, DLL + PWM modulators (Controller.qsch)



SHUFFLER 1 - converter between PV1+ (T) and PV3+ (B) with the midpoint on PV2+ (M): the hand-drawn v3 power stage

Leg A = M3 / M4 (driver X6, PWM_AH / PWM_AL), leg B = M1 / M2 (driver X7, PWM_BH / PWM_BL); X4 = ERUC23 inverse-coupled inductor.
Board capacitors C1 / C3 (130 uF) and C5 (225 nF); V13 = 0 V midpoint sense; V4 (A) and V7 (B) = 0 V sources in the windings, positive into the midpoint.
Ports: T / M / B = the converter rails (cable and copper inductances are on the top sheet); PWM_* = 3.3 V logic from the Controller block;
HS_A / HS_B = TMCS1126 hall-sensor outputs to the Controller (1.65 V + 100 mV/A on A, 1.65 V - 25 mV/A on B: reversed as mounted on the board).
Measure the winding currents from the top sheet as I(V4•XS1) and I(V7•XS1); their difference is the circulating current, their sum the shuffled current.



TMCS1126 HALL SENSORS - on the board at the inductor; 1k + 318.3p = the 500 kHz output pole

PARAMETERS - edit values here (units: m = milli, u = micro, n = nano, p = pico)

```
* PWM
.param Tsw=5u                  PWM period: 5 u = 200 kHz (FSWP for the DLL follows 1/Tsw)
.param Tdead=200n              dead time cut from both edges of every FET pulse
.param Vlogic=3.3              logic level of the PWM into the gate drivers
* controller enable and open-loop duties
.param TEN=1m                  controller enable time: open loop before it, closed loop after
.param D0A1=0.5 D0B1=0.5 D0A2=0.5 D0B2=0.5   duties before TEN (leg A / B of converter 1 / 2)
* reference steps
.param VSTEP1=0 VSTEP2=0 TSTEP=10m   voltage-difference references (VM1-VM2, VM2-VM3) stepped at TSTEP
* DLL test inputs
.param KSCALE=0.3              gain scale; 0.3 = firmware boost default (kscale in 7_PI/cod/params.yaml)
.param MSEL=0                  0 = fil compiled into the DLL (params.h), 1 = the 100 kHz 5th M, 2 = ctrl_M2.txt
.param FILT=0 FC=2000          0 = first-order IIR at FC Hz; 1..8 = moving average over that many control ticks
.param NOISE=0                 sensor noise: 1 = bench level
.param SAMP=1                  1 = firmware hall sampling (4 samples per PWM period), 0 = ideal average, 2 = top of triangle
.param KSERR=1                 B-phase gain error factor
.param VM1SRC=0 VSTRP=0        VM1 source (0 = INA239); VSTRP string-voltage option
* kept from the drawn v3 sheet
.param VT=0.009                bottom FET on while D < RAMP < 1 - Tdead/Tsw
```

SIMULATION

```
.tran 20m                      the 15 ms shade ramp of V1 / V2 starts inside the window
.options timer = 1n
.plot I(I1)*V(Out)
* save keeps the .raw small: every node of the measurements, add what you want to plot
.save V(Out) V(PV2P) V(PV3P) V(T1) V(B1) V(HS_A1) V(HS_B1) V(HS_A2) V(HS_B2) I(I1)
.save V(PWM_AH1) V(PWM_BH1) V(PWM_AH2) V(PWM_BH2) I(V4•XS1) I(V7•XS1) I(V5•XS2) I(V6•XS2)
.save V(DA1•XC) V(DB1•XC) V(DA2•XC) V(DB2•XC) V(STRESS•XC) V(BETA•XC) V(RV1•XC) V(RV2•XC)
* functions used inside the sub-sheets (a top-level .func is visible there)
.func hall_cur(_gain) = 1.65+gain*cur
.func duty(en,cmd,dtf) = if(en>0.5,cmd,dtf)
.func pwm_top(rmp,dty) = if(time<1n,0,if(rmp<dty-Tdead/Tsw,Vlogic,0))
.func pwm_bot(rmp,dty) = if(time<1n,0,if(rmp>dty,if(rmp<1-Tdead/Tsw,Vlogic,0),0))
* TMCS1126 output: 1.65 V + gain * winding current
* commanded duty: the DLL output when enabled, else the open-loop duty
* top if rmp < RAMP < D - Tdead/Tsw (all low at t = 0)
* quick check: .tran 1.5m with TEN=0.3m VSTEP2=1 TSTEP=0.8m (meas window 1.3m-1.5m) solves in about 30 s
* usual next move: .step param KSCALE list 0.3 1
```

MEASUREMENTS - settled window 18-20 ms of the 20 ms run

```
.meas tran vd1 AVG V(Out,PV2P)-V(PV2P,PV3P) FROM 18m TO 20m
.meas tran vd2 AVG V(PV2P,PV3P)-V(PV3P) FROM 18m TO 20m
.meas tran ic1 AVG I(V4•XS1)-I(V7•XS1) FROM 18m TO 20m
.meas tran ic2 AVG I(V5•XS2)-I(V6•XS2) FROM 18m TO 20m
.meas tran sh1 AVG I(V4•XS1)+I(V7•XS1) FROM 18m TO 20m
.meas tran sh2 AVG I(V5•XS2)+I(V6•XS2) FROM 18m TO 20m
.meas tran da1 AVG V(DA1•XC) FROM 18m TO 20m
.meas tran db1 AVG V(DB1•XC) FROM 18m TO 20m
.meas tran da2 AVG V(DA2•XC) FROM 18m TO 20m
.meas tran db2 AVG V(DB2•XC) FROM 18m TO 20m
.meas tran pout AVG I(I1)*V(Out) FROM 18m TO 20m
.meas tran vout AVG V(Out) FROM 18m TO 20m
.meas tran iout AVG I(I1) FROM 18m TO 20m
.meas tran stressmax MAX V(STRESS•XC)
.meas tran hsa1 AVG V(HS_A1) FROM 18m TO 20m
.meas tran hsb1 AVG V(HS_B1) FROM 18m TO 20m
.meas tran hsa2 AVG V(HS_A2) FROM 18m TO 20m
.meas tran hsb2 AVG V(HS_B2) FROM 18m TO 20m
.meas tran pwmah1 AVG V(PWM_AH1) FROM 18m TO 20m
.meas tran pwmah2 AVG V(PWM_AH2) FROM 18m TO 20m
* vd1 / vd2: module voltage differences the controller runs; ic / ishc circulating / shuffled
* inductor currents of each converter; da / db: commanded duties; pout / vout / iout: string power, voltage, current;
* stressmax: supervisor peak stress; he*: hall outputs (1.65 V at zero current); pwmah*: top-FET PWM average
```

EPISODE V:
QSPICE STRIKES BACK

CONTROLLER - shuffler_ctrl_dll2.dll: the firmware control stack (6_RTOS/ctrl_core) as a QSPICE C-block

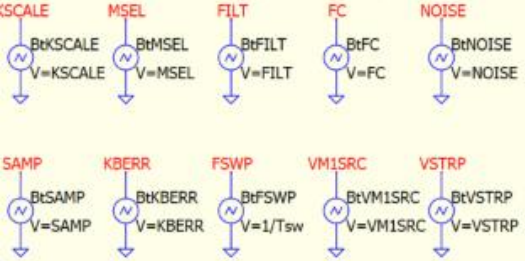
The DLL models the INA239 / hall sampling and quantisation, the 10 kHz control tick and the gain matrix M from params.h (MSEL = 0), the 100 kHz S9i M (MSEL = 1) or ctrl_M2.txt (MSEL = 2). shuffler_ctrl_dll2.dll and ctrl_M2.txt must sit next to the top sheet.
Open loop until TEN (CTRL_EN low: the duties D0A1 .. D0B2), closed loop after. Every test input of the DLL is a .param on the top sheet.
Inputs: PV1P / PV2P / PV3P = the string nodes the INA239s see, HS_* = hall outputs of the shufflers. Outputs: one PWM pair per leg.
Signal flow left to right: sources -> DLL -> duty override (duty() .func) -> PWM_Modulator XP1 (converter 1) and XP2 (converter 2).

ENABLE, REFERENCE STEPS AND PWM RAMPS

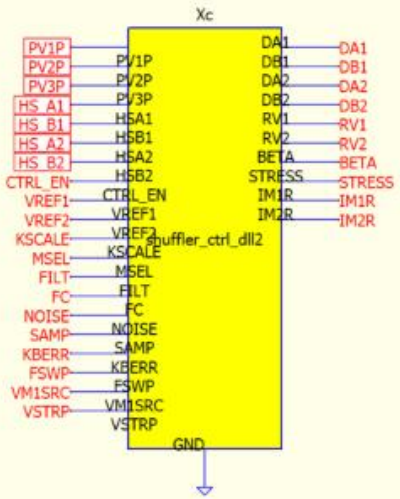




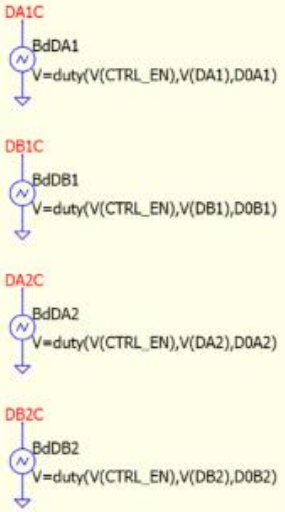
DLL TEST INPUTS - each a .param of the top sheet



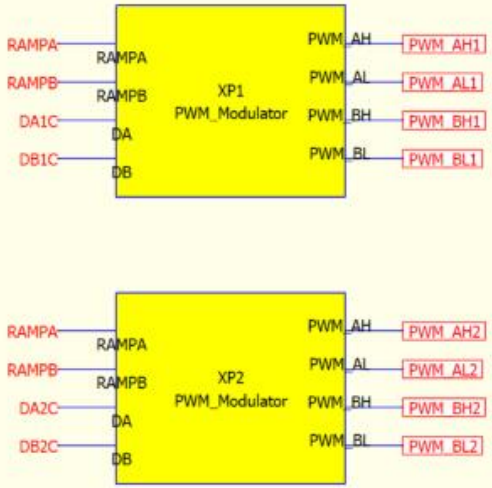
shuffler_ctrl_dll2.dll - INA239 / hall sampling, 10 kHz control tick, M from params.h



COMMANDED DUTIES - DLL output when CTRL_EN, else D0*

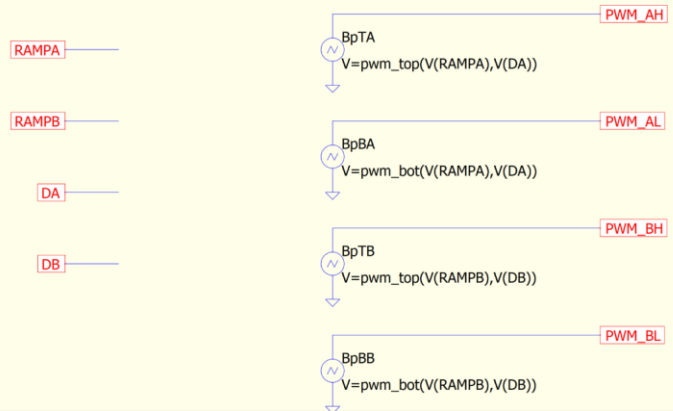


PWM MODULATORS - one per converter (hierarchical sheet PWM_Modulator)

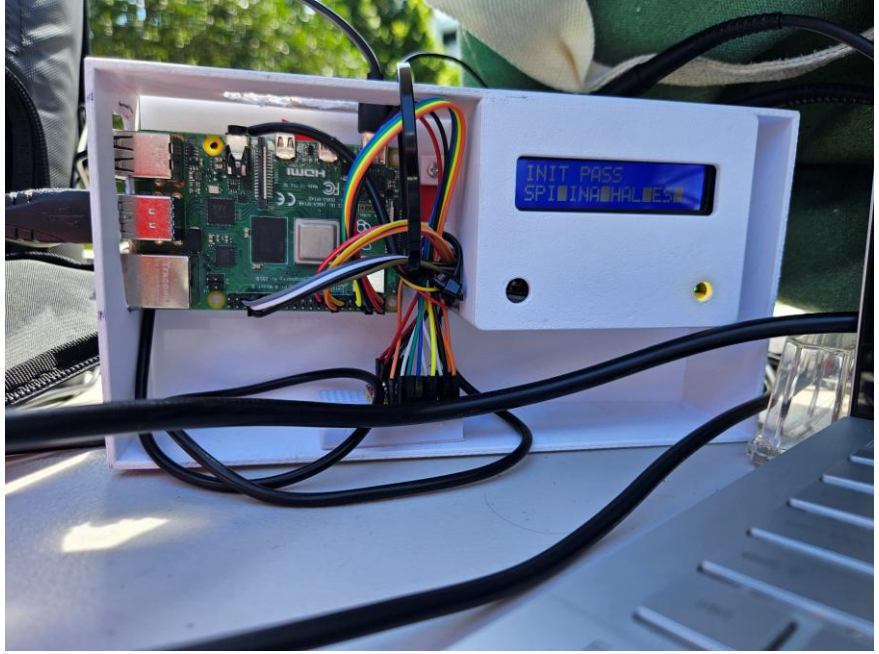


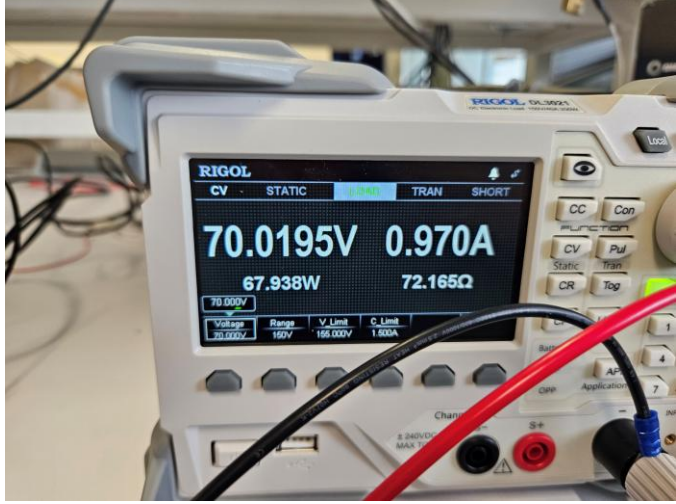
PWM MODULATOR - HRTIM-style comparators of one converter

Top FET on while RAMP < D - Tdead/Tsw, bottom FET on while D < RAMP < 1 - Tdead/Tsw: Tdead is cut from both edges of every pulse.
RAMPA / RAMPB are the 0..1 sawtooth ramps of leg A and leg B (leg B half a period later), DA / DB the commanded duties of the legs.
All outputs are low for t < 1 ns so the operating point is found with the FETs off (pwm_top / pwm_bot are .func on the top sheet).

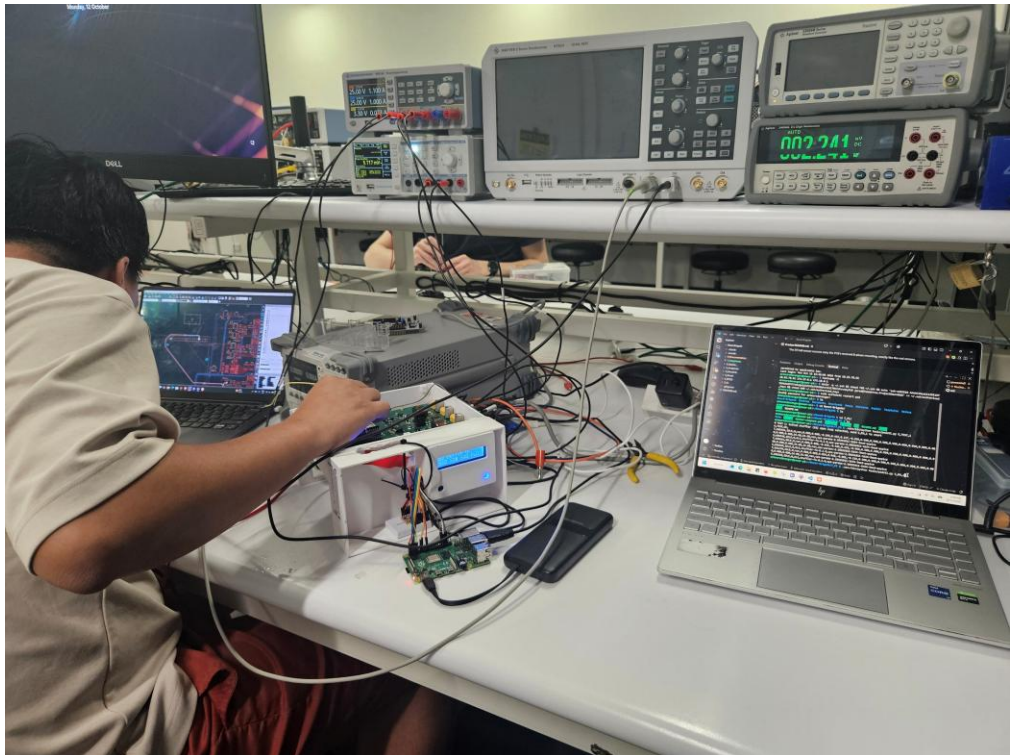


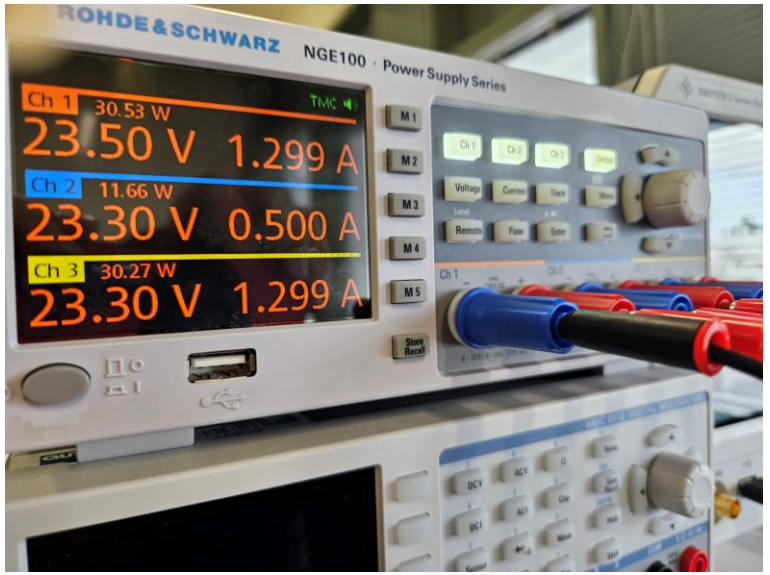










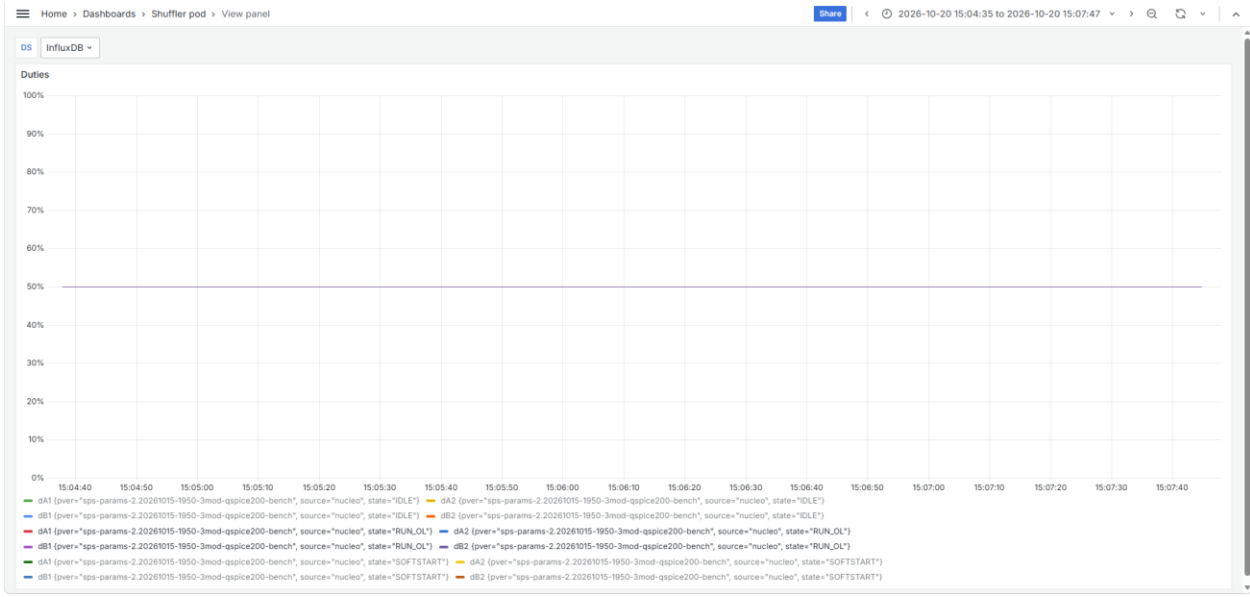


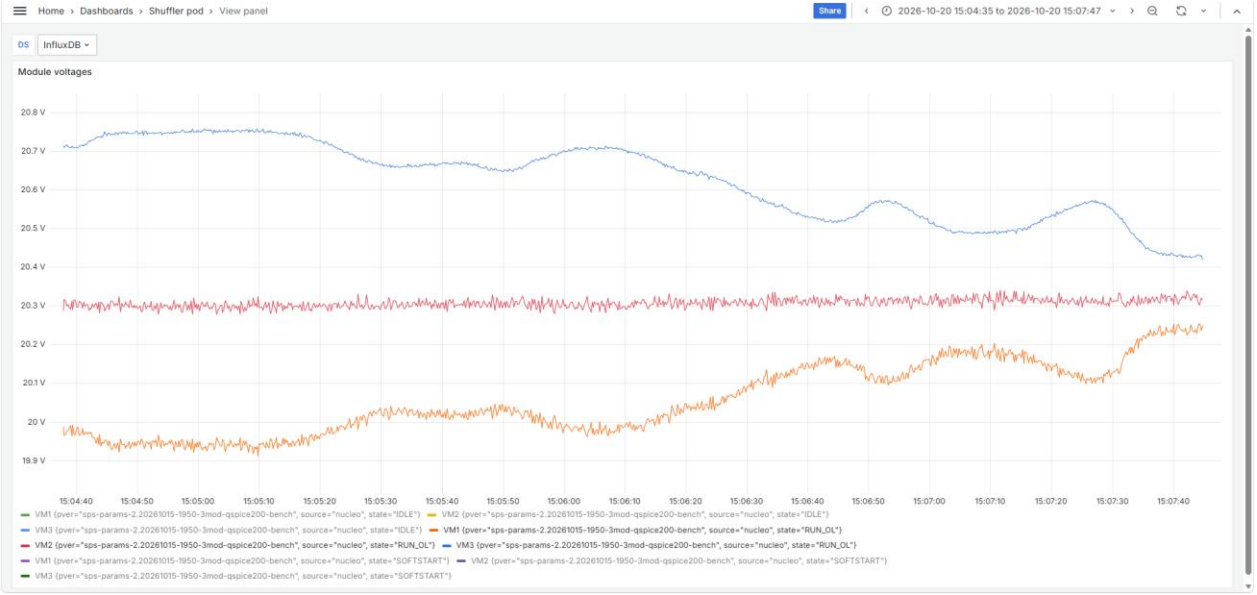
EPISODE VI:
RETURN OF THE SHUFFLER

Solar Test Day (20-10-2026)

200 kHz Open Loop Peak Sun









Solar Test Day (20-10-2026)

200 kHz Closed Loop Peak Sun
kscale = 10%





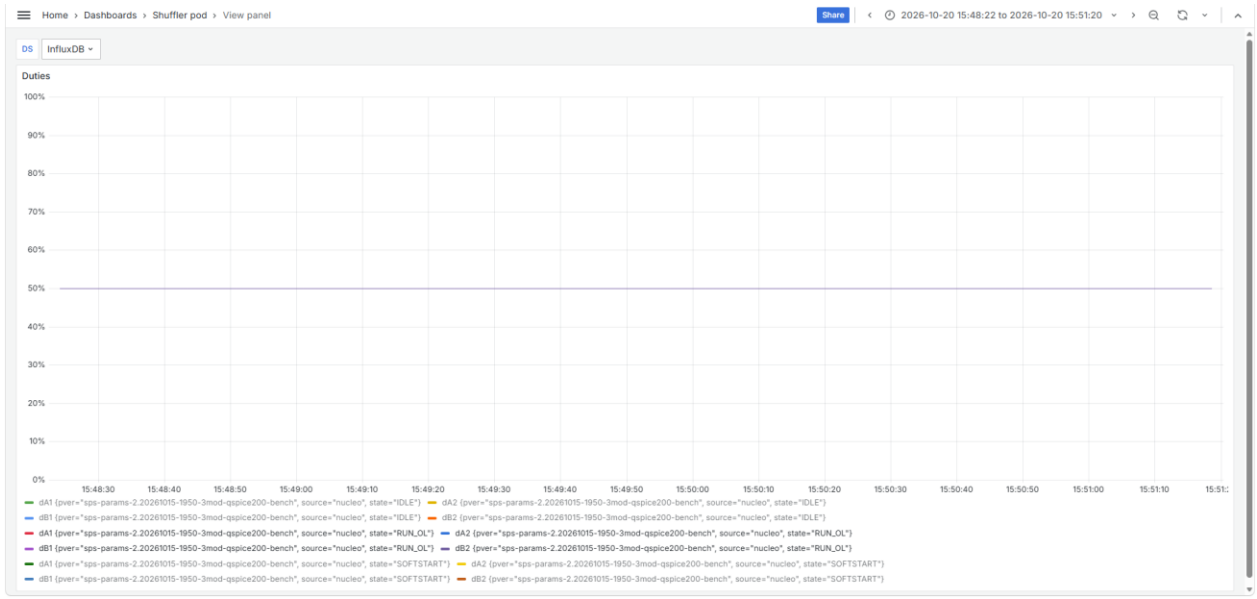


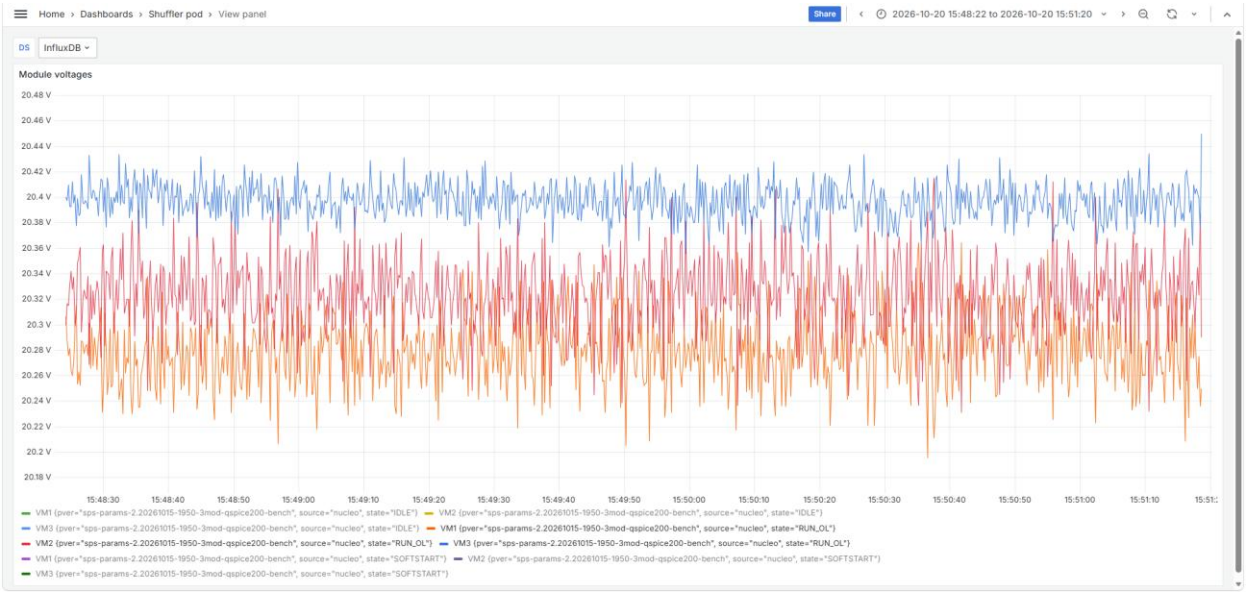


Solar Test Day (20-10-2026)

200 kHz Open Loop Minimal Sun







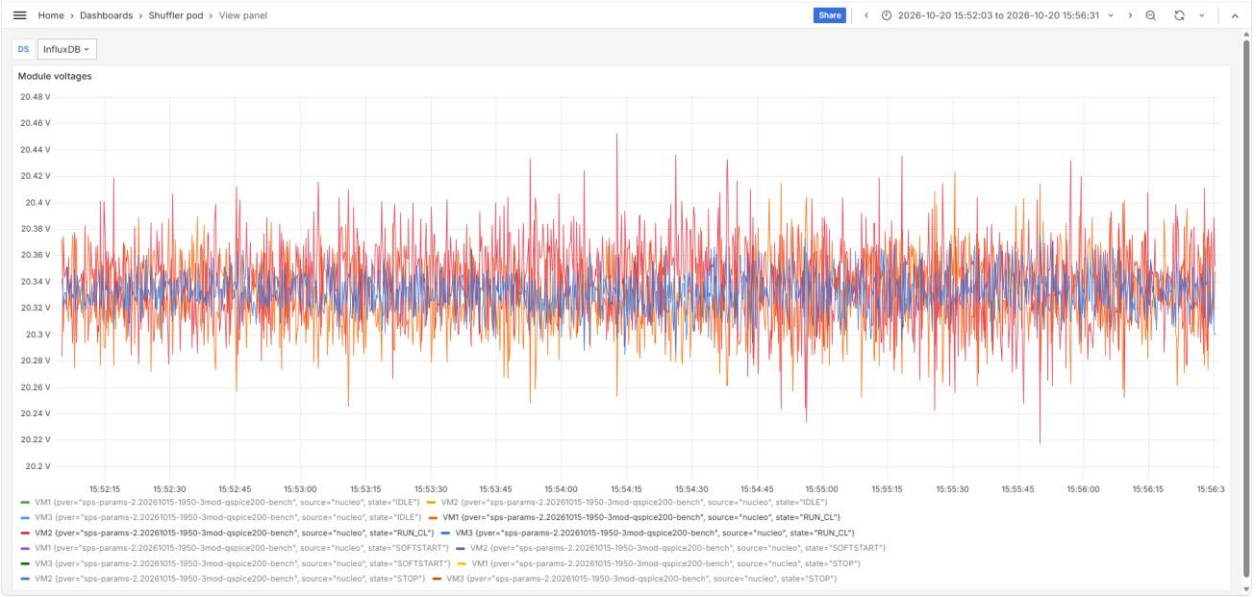


Solar Test Day (20-10-2026)

200 kHz Closed Loop Minimal Sun
Kscale = 10%










DIGITECH
Autoranging
True RMS

Blueprint for Scaling Up

Mechanical:

- Enclosure Design
- Thermal Analysis

Electrical:

- Decouple the 2x shufflers
- Provision for a chain of boards
- Centralised controller board (master-slave approach)

Software:

- Many (‘N’) shuffler control
- Analyse how further coupling affects dynamics / control
- Multiple strings (array)
- Integrate MPPT into the system.

* Problem Identification
Re-evaluate target DPP application.
Is solar going to have the most value?

Risks

- Scope Creep
- Firmware Failure (E-STOP and State Telemetry)
- Safe Outdoor Testing (Sun, Fire, Rain)
- Electrical Safety (ratings, short circuits)
- Data Security
- Inaccurate model
- Controller Faults
- Sensor Faults

Ethics & Sustainability

Ethics

- Electrical Safety
- Data Security / Privacy
  - Authentication via Raspberry Pi (locally hosted)
  - Only streams sensed data and internal states
- Integrity

Sustainability

- Work performance through constant collaboration
- Longevity of the project through docs, videos and github
- Enclosure / Thermal considerations

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Appendix Contents

- A: Timeline
- B: The Journey (Fun Photos)
- C: Solar Test Raw Data Minimal Sun
- D: Bench Test Raw Data
- E: Analytical Solution (Single Shuffler / Lead Up)

Appendix A: Timeline

Semester 1

- Problem Identification – Wk 4
- Existing DPP / MPPT Control Strategies - Wk 7
- Analytical Solution using classical methods – Wk 10
- Simulink Plant Model – Wk 13

Break

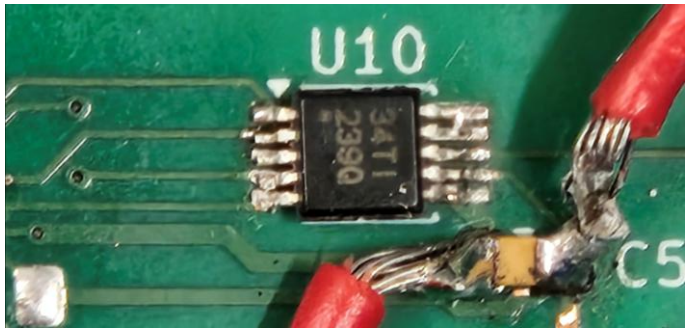
- Analytical
- Model-in-the-loop
- Software-in-the-Loop

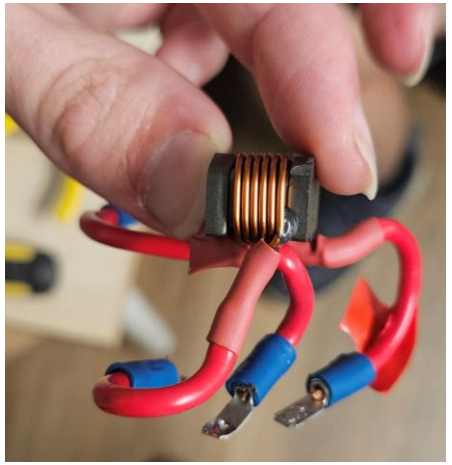
Semester 2

- Component Selection - Wk 4
- PCB Order – Wk 9
- Enclosure Development – Wk 10
- Firmware / Software – Wk 11
- Theoretical Integration / Gain Refinement – Wk 12
- Bench Test – Wk 12
- PV Experiment – Wk 13
- Post-Processing – Wk 13

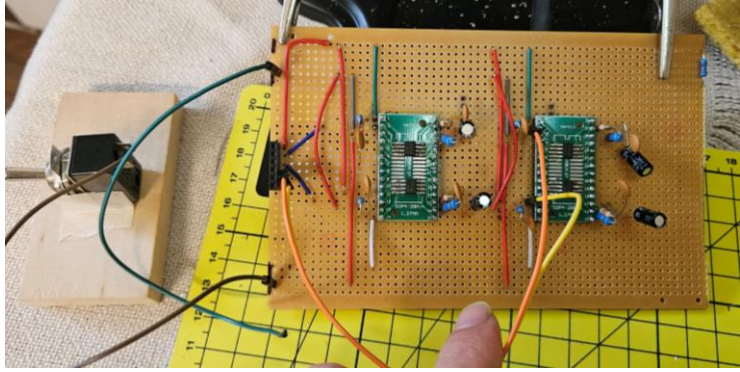
* Had to compromise on pcb revisions and features

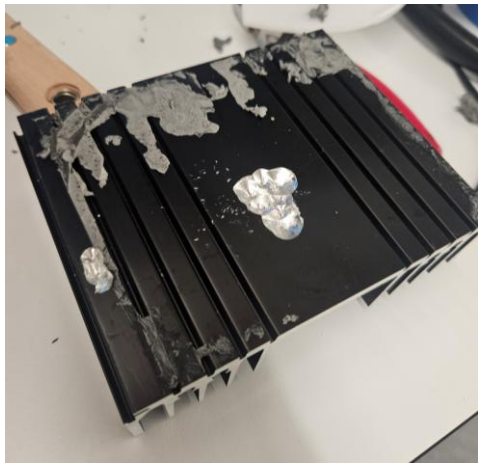
Appendix B: The Journey



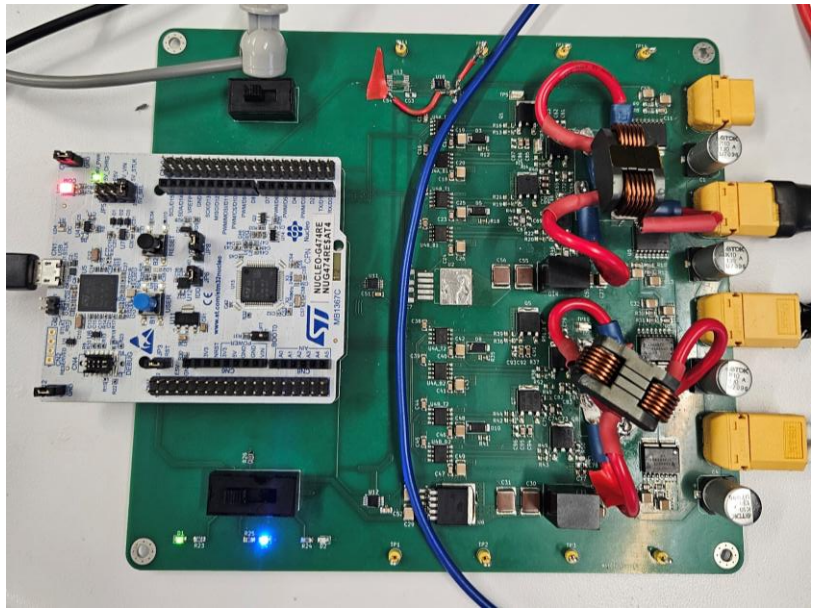


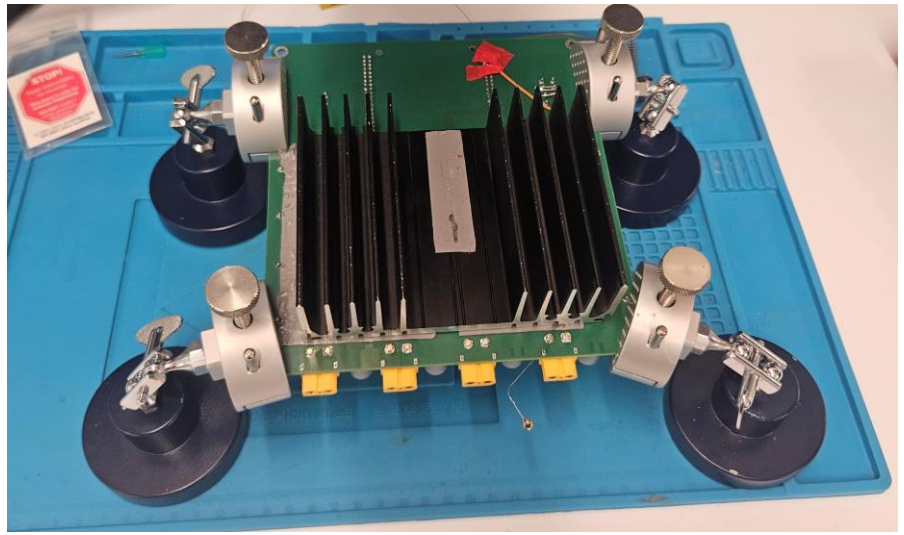










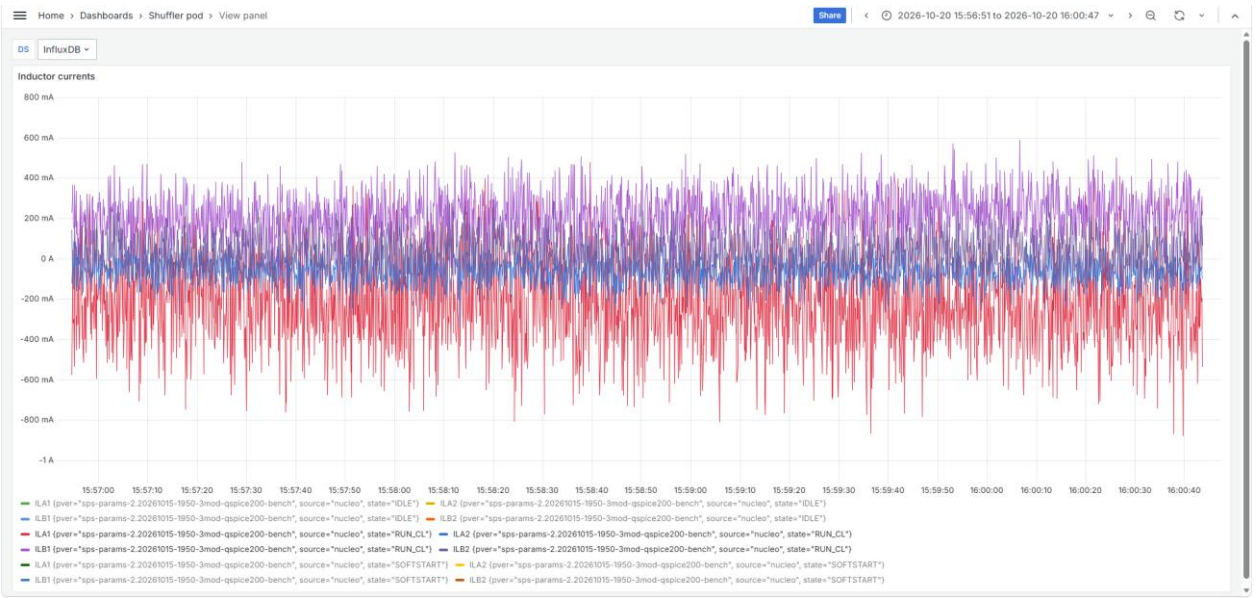


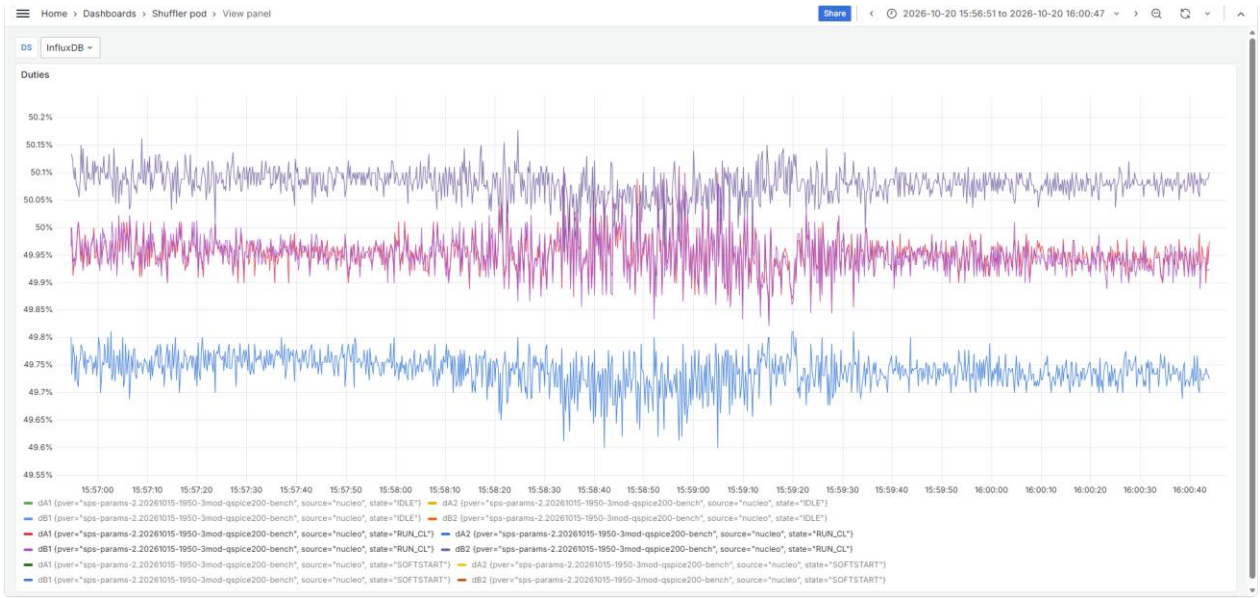
Appendix C: Solar Test Raw Data Minimal Sun

- 200 kHz CL : {20%, 30%, 40%, 50%}

Solar Test Day (20-10-2026)

200 kHz Closed Loop Minimal Sun
Kscale = 20%



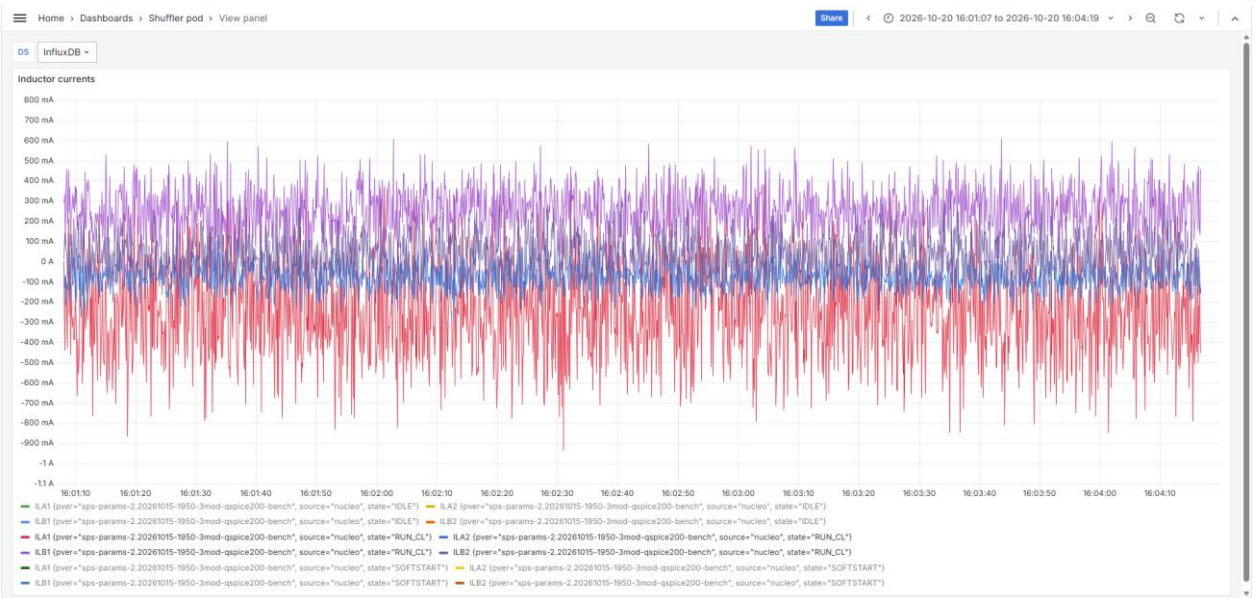






Solar Test Day (20-10-2026)

200 kHz Closed Loop Minimal Sun
Kscale = 30%







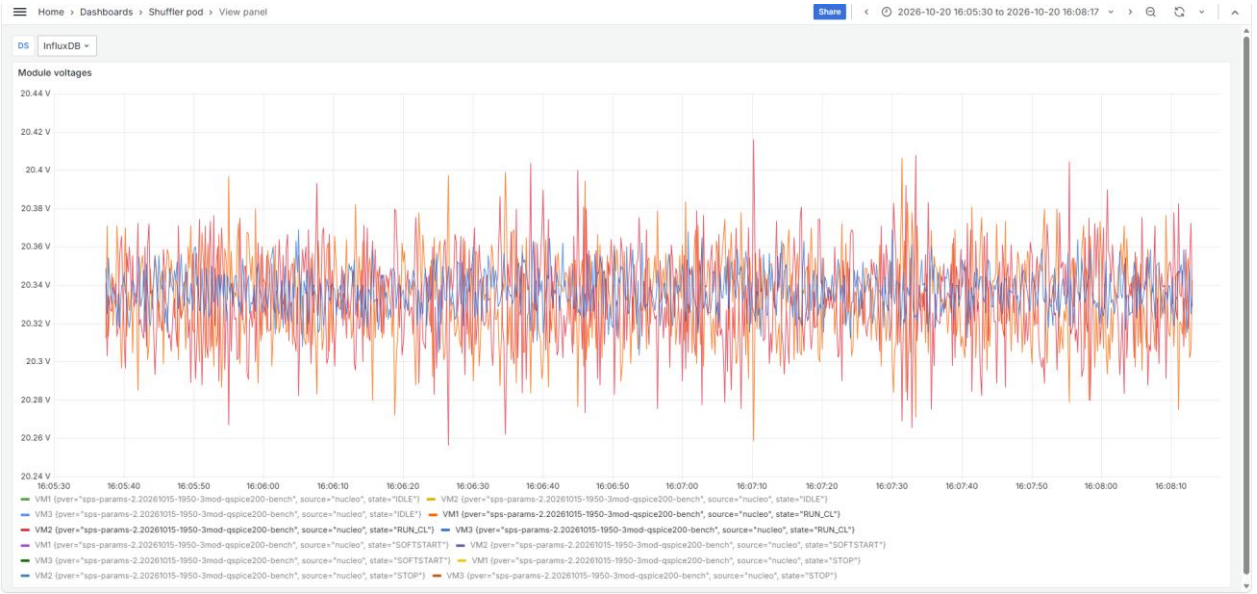


Solar Test Day (20-10-2026)

200 kHz Closed Loop Minimal Sun
Kscale = 40%



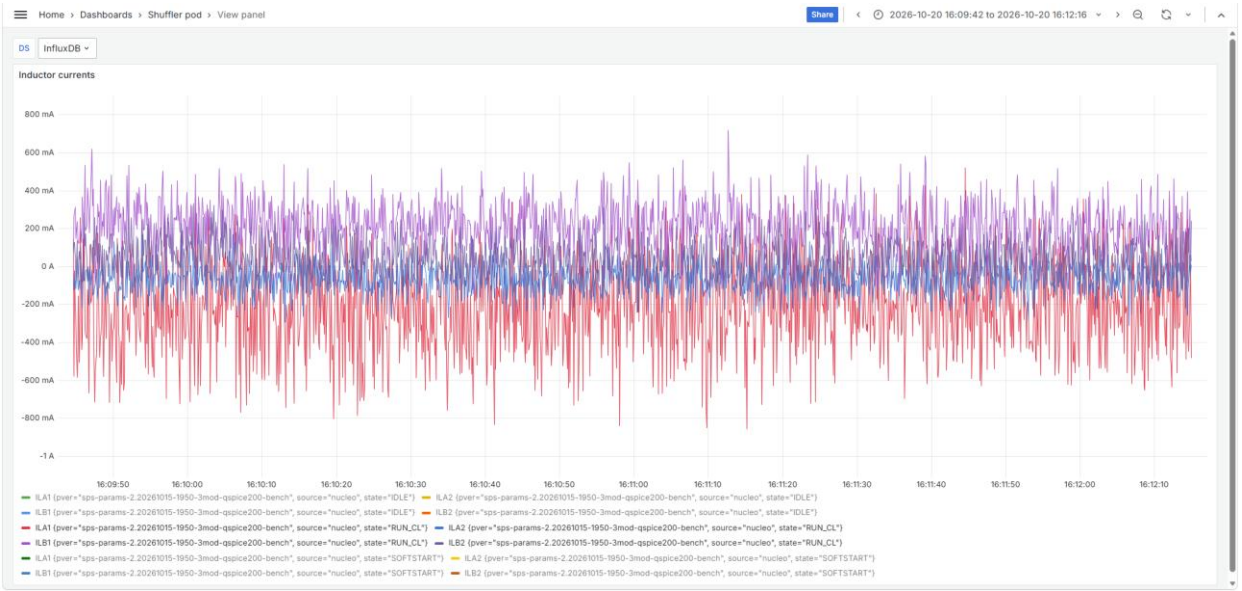






Solar Test Day (20-10-2026)

200 kHz Closed Loop Minimal Sun
Kscale = 50%









Appendix D: Bench Test Raw Data

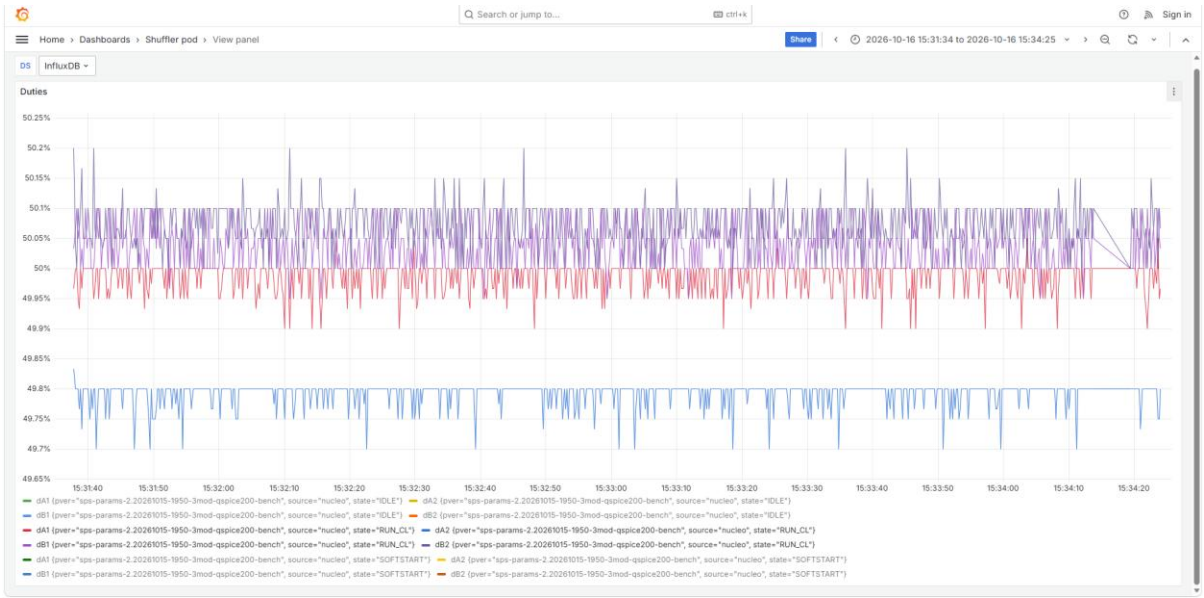
- 200 kHz {OL, 10, 20, 30}
- 300 kHz {OL, 30, 50}
- 400 kHz {OL, 30}

Bench Test Day (16-10-2026)

200 kHz kscale = 10%

- Before ADC Fix
- PSU was still freaking out







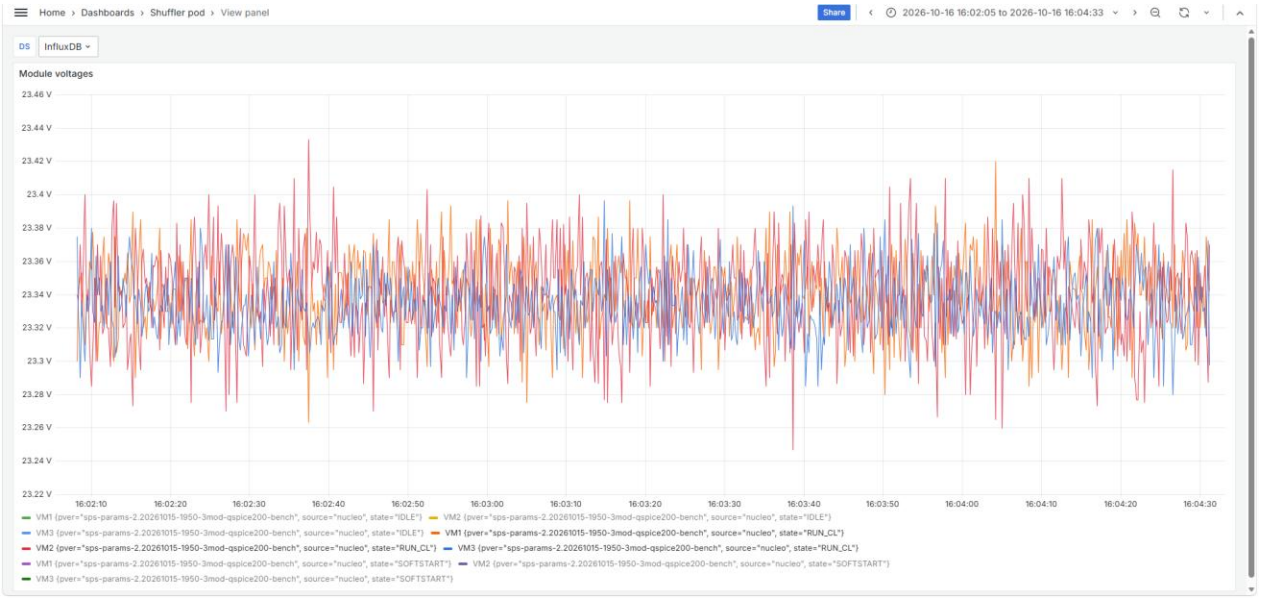


Bench Test Day (16-10-2026)

200 kHz kscale = 20%
* After ADC Fix





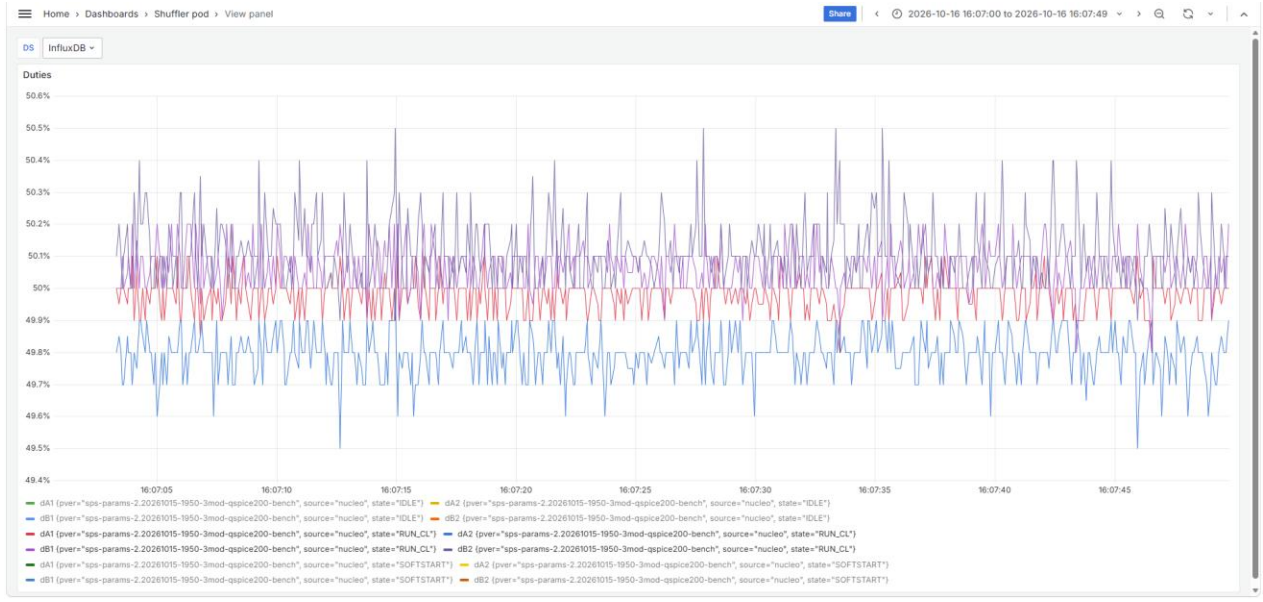




Bench Test Day (16-10-2026)

200 kHz kscale = 30%









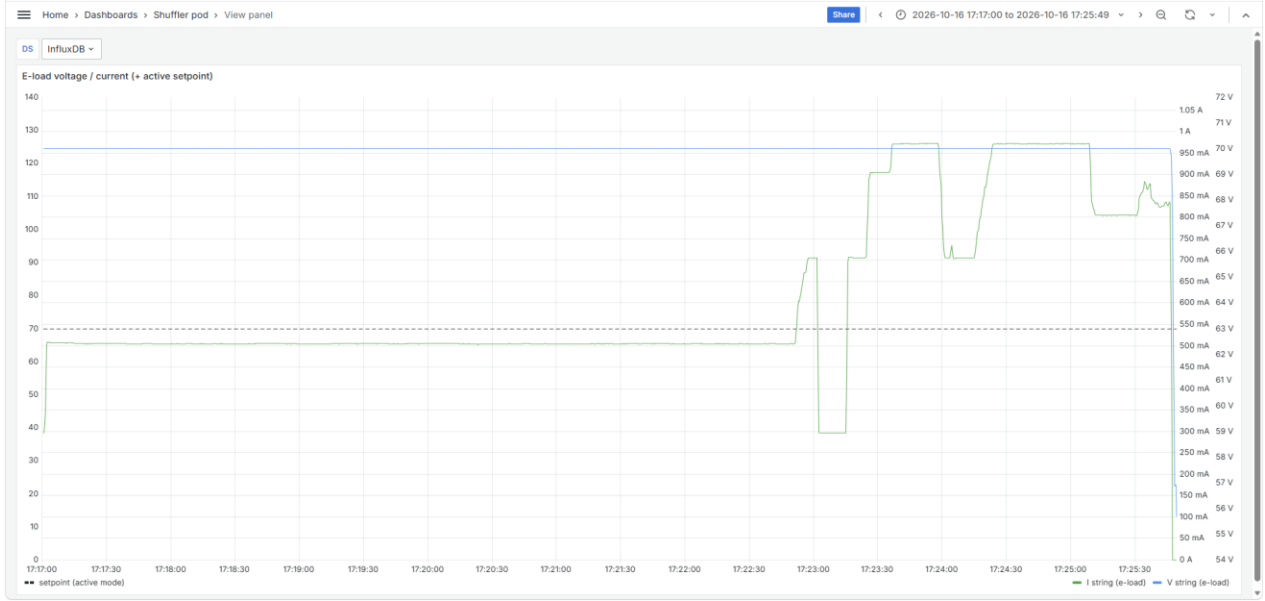
Bench Test Day (16-10-2026)

300 kHz kscale = 30%



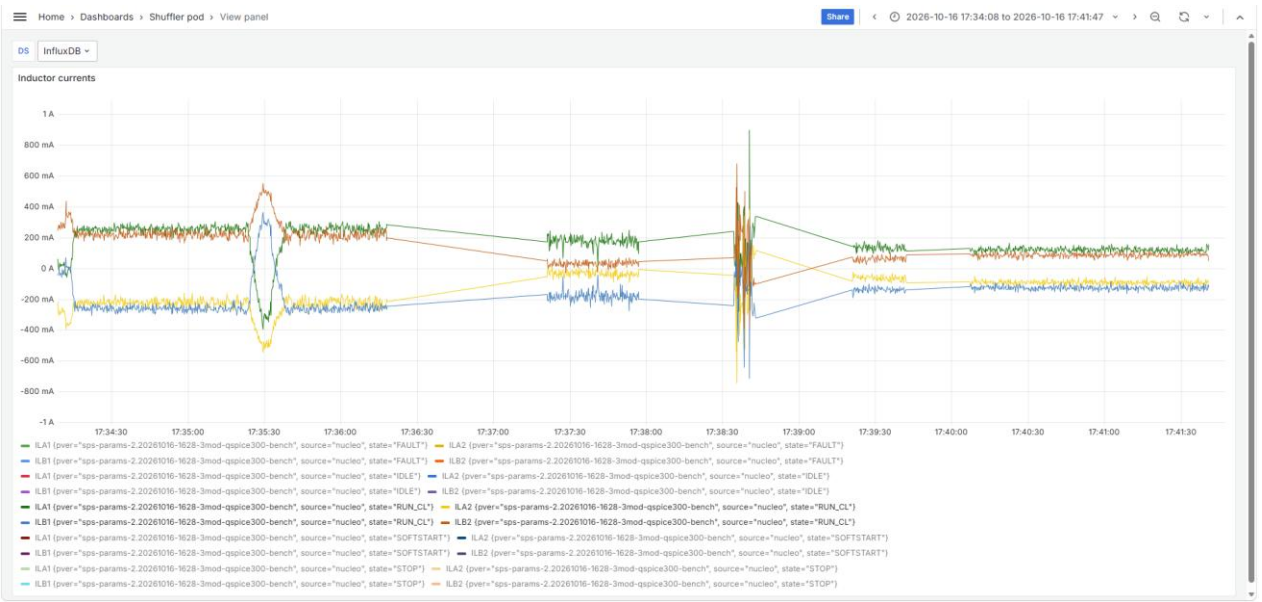






Bench Test Day (16-10-2026)

300 kHz kscale = 50%









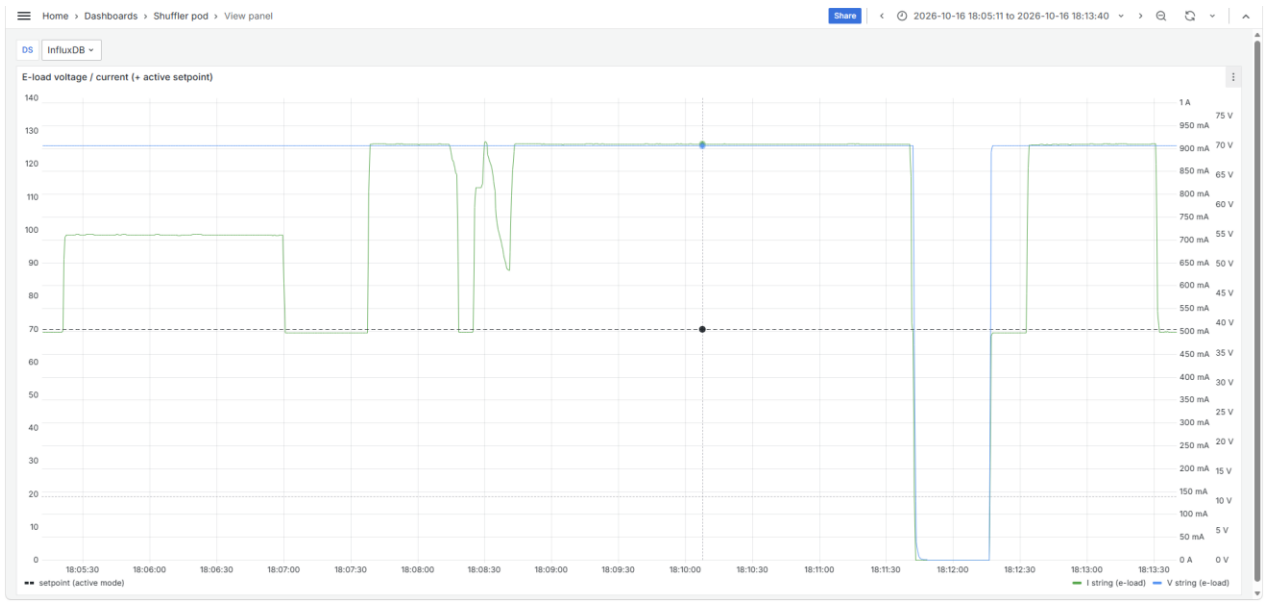
Bench Test Day (16-10-2026)

400 kHz kscale = 30%





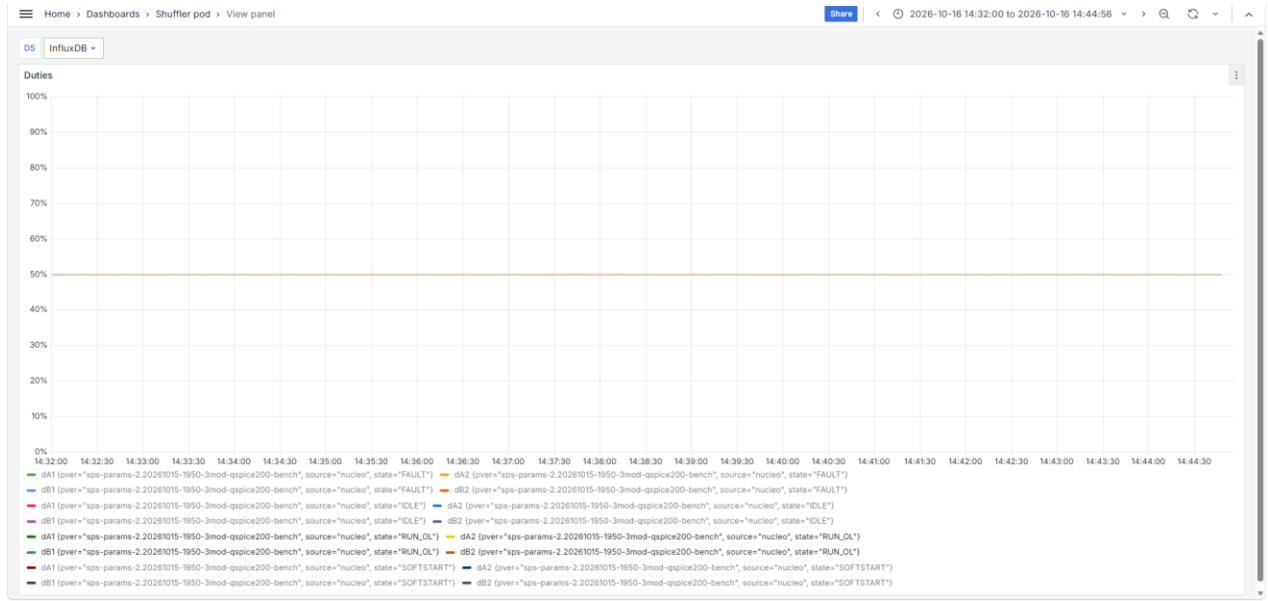


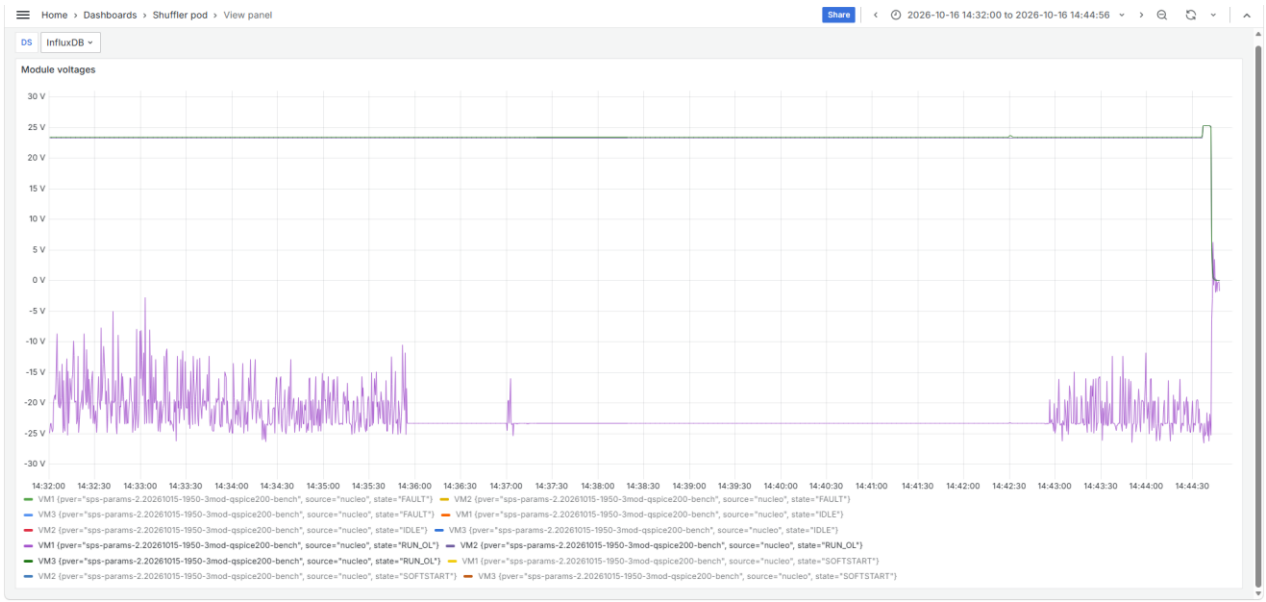


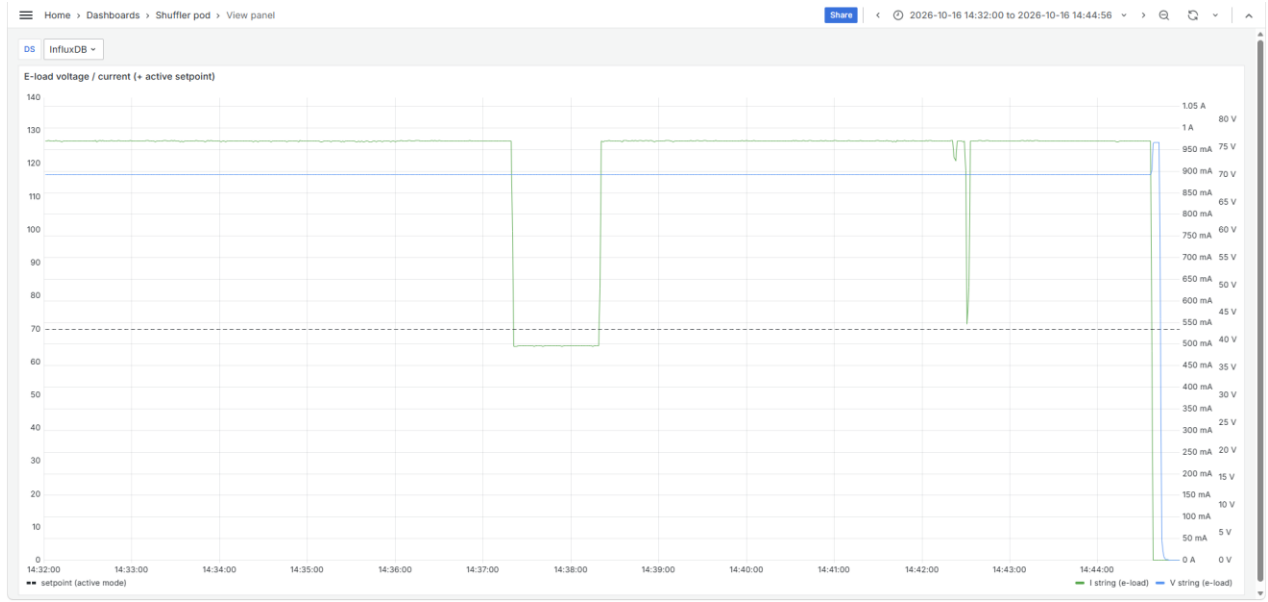
Bench Test Day (16-10-2026)

200 kHz Open Loop
* VM1 before recalculating with Eload





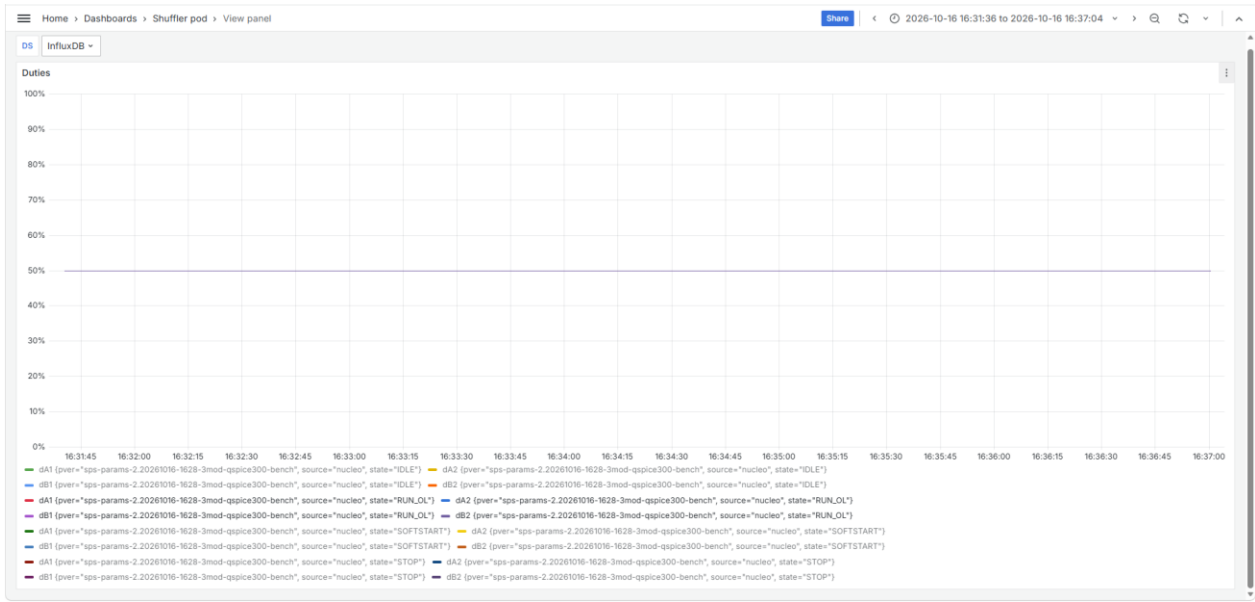


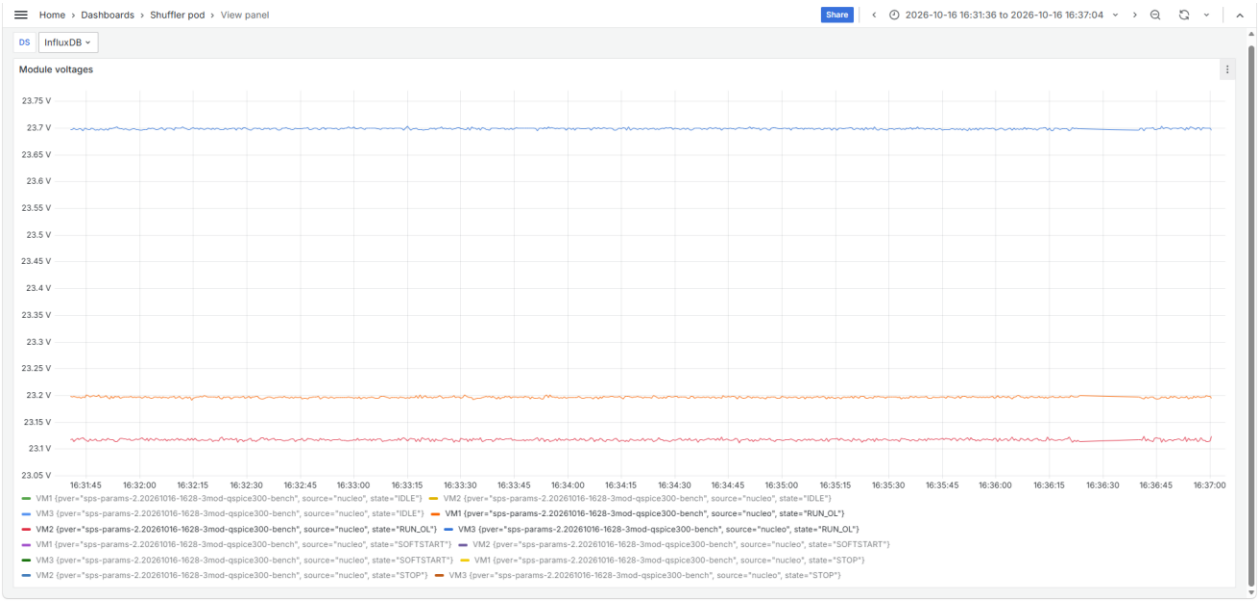


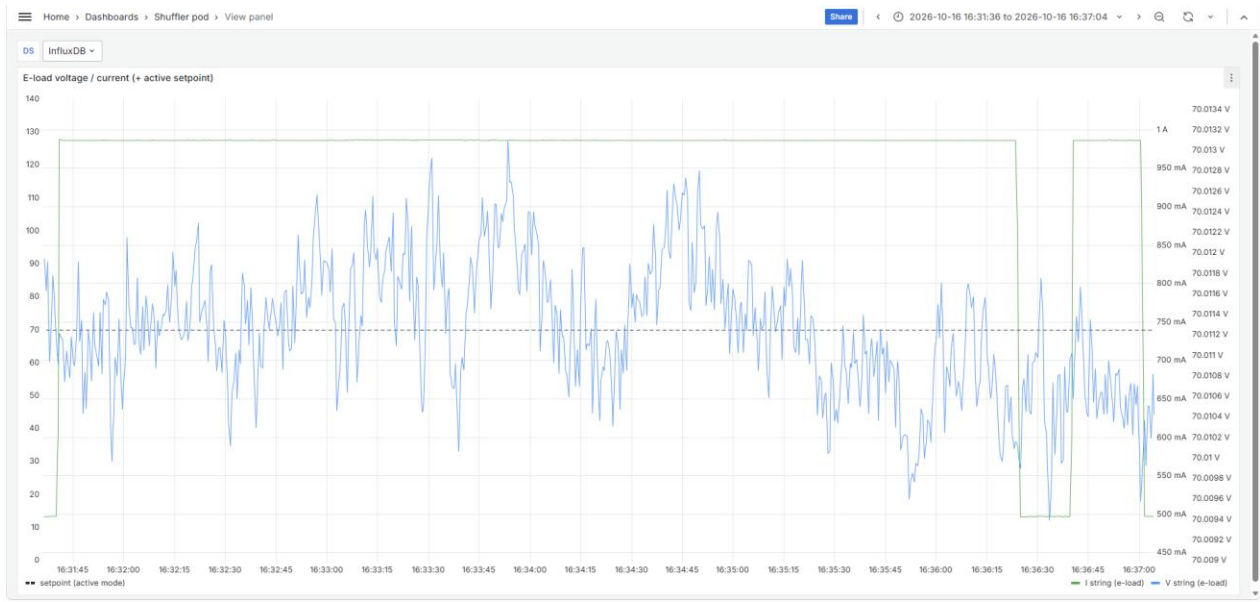
Bench Test Day (16-10-2026)

300 kHz Open Loop





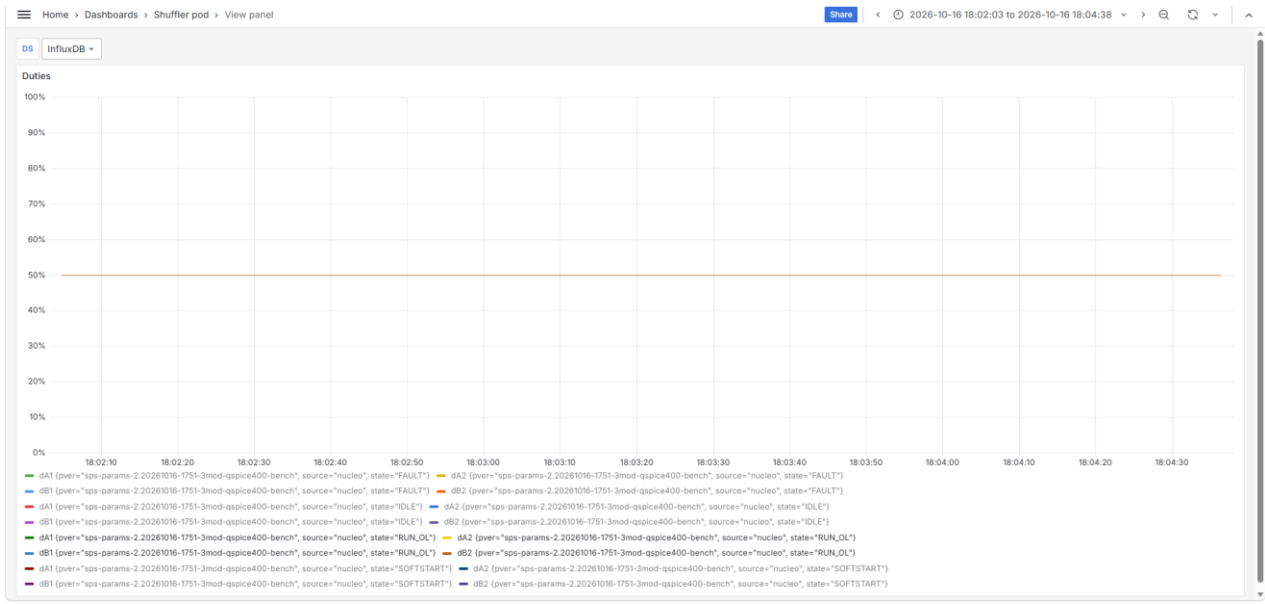




Bench Test Day (16-10-2026)

400 kHz Open Loop

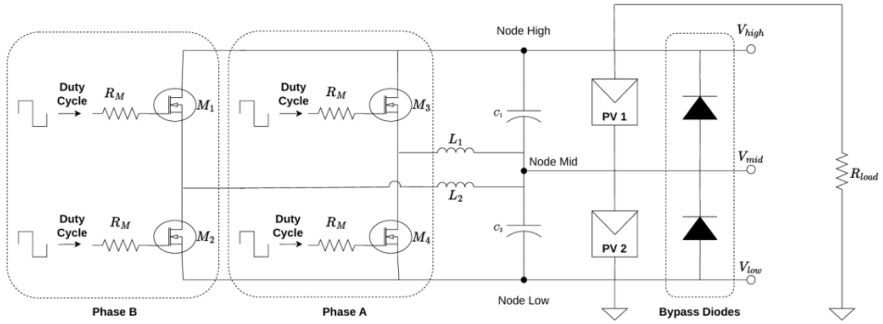








Appendix E: Analytical Solution (Early Stages)



- Single Shuffler Avg Model

State Vector: $x = [I_{L1}, I_{L2}, V_{C1}, V_{C2}]^{\mathrm{T}}$

Control Input Vector: $u_d = [d_A, d_B]^T$

Exogeneous Input Vector: $u_e = [I_{PV1}, I_{PV2}, I_{LOAD}]^T$

Equilibrium Point: the duty cycles of both phase legs must be equal at steady state. Equilibrium Duty (D).

$$\frac{\bar{V}_{C2}}{\bar{V}_{C1}} = \frac{D}{1-D} \quad ; \quad \mathrm{D} = \frac{I_{LOAD} - I_{PV1}}{I_{LOAD} - I_{PV1} + I_{PV2}}$$

EPISODE I:
THE ANALYTICAL MENACE

Analytical Solution – Linearisation

Common-mode duty: u = (dA + dB) / 2 => I_sum = IL1 +IL2

Differential-mode duty: u = dA – dB => I_diff = IL2 – IL2

Marginal Stability: Poles on the imaginary axis

Natural Resonance: approx. 5.5 kHz

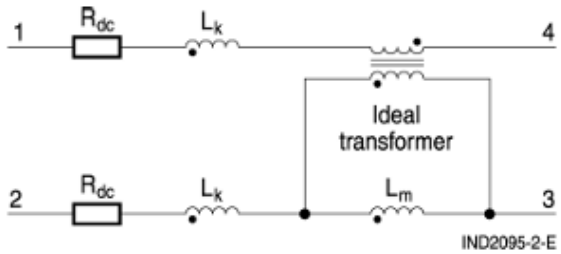
* This is why my early control designs didn’t work. Without parasitics the model is an undamped oscillation.

Analytical – Controllability & Observability

- Differential Mode is controllable if $V_{sum} \neq 0$
  - Active bus
- Common Mode is controllable if $I_{sum} \neq 0$
  - Converter doesn't enter DCM (ie. Avg inductor current is 0)
- Duty cycle is not saturated D = {0,1}

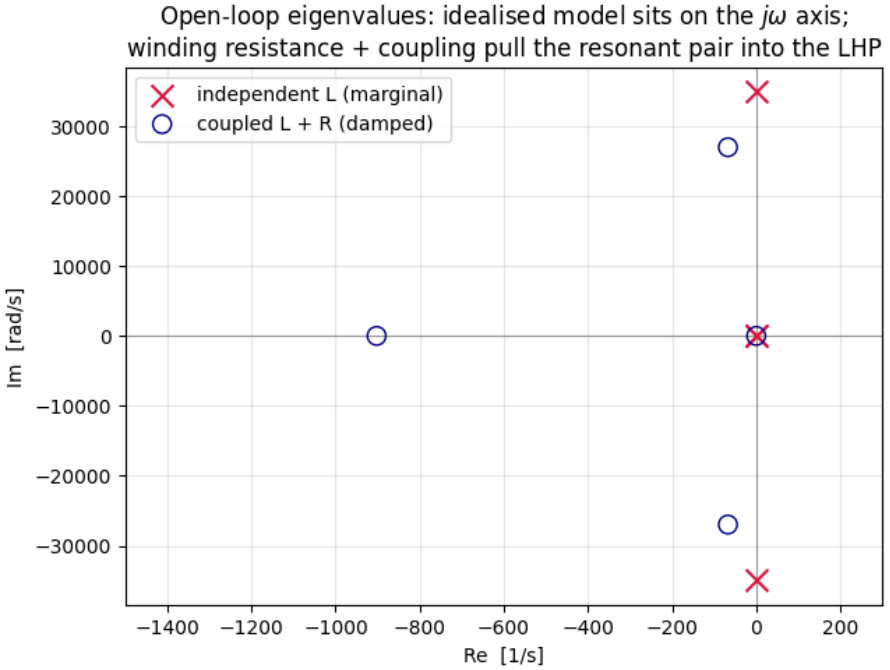
- Full State Feedback required for both modes. Voltage sensing provides common mode. Need current sensors for each phase to use differential mode.

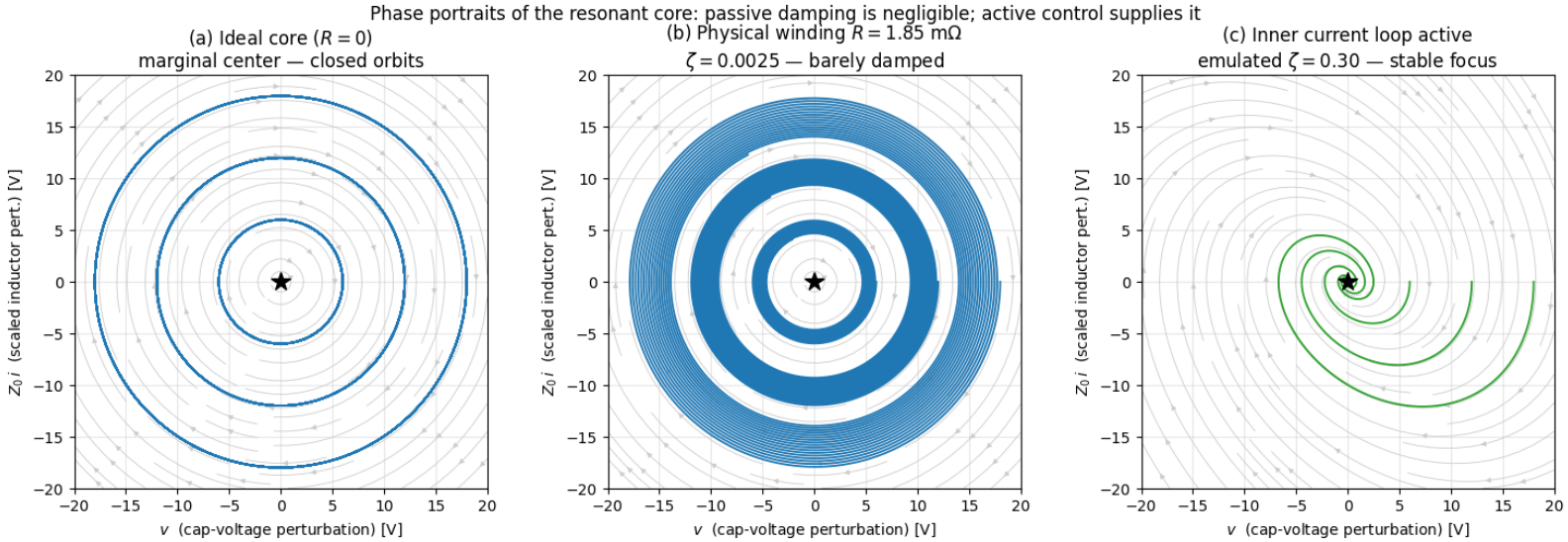
Coupled Inductor vs Independent Inductors



$L_k = 0.5 \bullet L_3$
$L_m = L_2$ (or L_1) $- 0.5 \bullet L_3$
Example: ERUC23-4R1K (4.1 µH)
R_{DC} = 1.85 mΩ
L_k = 2.05 µH
L_m = 5.85 µH

[11]





Change of Thinking

- The problem with the previous control model is that it didn't acknowledge the coupling between the states. It treated everything too independently which led to a 3 layers of SISO feedback loops.

- MIMO LQR and Integral Regulator

